The logic of doxastic grading: draft

1 Introduction

In formal epistemology, three ways of representing agents' beliefs are widely discussed:

1. Categorical belief: each proposition is either believed, or its negation is believed, or neither of the two (Hintikka 1962; Fagin et al. 2004; Baltag and Smets 2006; Van Ditmarsch et al. 2015);
2. Probabilistic belief: each proposition is ascribed a real number between 0 and 1 that expresses the agent's estimate of the proposition's likelihood (Ramsey 1926; Savage 2012; Jeffrey 1990);
3. Relative belief: propositions/possible worlds are getting ordered by how plausible the agent finds them to be (Halpern 1997; Segerberg 1971).

All three options come with widely accepted sets of rationality norms. A rational agent's categorical beliefs should agglomerate, be consistent, and be closed under logical entailment (Hintikka 1962; Kyburg 1961; Makinson 1965). E.g., it is irrational to believe that $p \wedge \neg p$, to believe that p and believe that q without believing that $p \wedge q$, or to believe that p and $p \rightarrow q$, while not believing that q. Rational probabilistic beliefs should be consistent with probability axioms (Kolmogorov 1950; Ramsey 1926). E.g. if an agent's credence in p is 0.2, then it is irrational for her to ascribe any credence to $\neg p$ other than 0.8. As for relative beliefs, a rational agent's strict plausibility order should be transitive and irreflexive (Halpern 1997, Chapter 2.7). E.g., no rational agent would believe that p is strictly more plausible than itself, or that p is strictly more plausible than q, q is more plausible than r, but p is not strictly more plausible than r.

Recently, the fourth way of representing beliefs is getting more attention, coming from the voting theory: qualitative grades. Each proposition gets graded on a qualitative (i.e., ordered, but not necessarily numeric) scale. The grade assigned to the proposition reflects the grader's assessment of how good a hypothesis it is, or how likely the proposition is to be true. Initially, qualitative grades were used to express agents' preferences, not beliefs (Balinski and Laraki 2007; Balinski and Laraki 2011), but the same approach is adopted for the epistemic case as well (Lyon and Morreau 2018). As of now, no set of rationality norms is accepted for epistemic qualitative grading. Moreover, in voting theory, where the graded objects are not assumed to be propositions, no such norms are imposed. Any grading should be possible: such a requirement is seen as an adaptation of the *unrestricted domain* principle with respect to qualitative grading.

In this paper, we show that this should not hold in an epistemic context. The connections between propositions create very natural rationality constraints on logically consistent grading. For instance, it is straightforward that p cannot be rationally graded strictly higher than $p \vee q$, as the former entails the latter; a logically inconsistent proposition cannot receive any positive grade at all, while a tautology should always be graded as high as possible.

The paper is organised as follows. In Section 2, we clarify our idealisations and conceptual grounds: how we understand evidence and evidential support of propositions, what qualitative grades we have in mind, and what justifies the agent's choice of grades. In Section 3, we present a technical background that we need to expose our formal model: some widely used techniques of representing evidential support for propositions via topological semantics for modal logic. In Section 4, we develop a logic of graded evidential support, where we can qualitatively reason about how reliable the evidence in support of each proposition is. In Section 5, we use this logic to derive the rationality norms for the qualitative grading of propositions. In Section 6, we study how qualitatively graded beliefs can be binarised, obtaining some negative results.

2 Conceptual grounds

We start by clarifying what doxastic grading is, and why it makes sense to represent an agent's beliefs that way.

Definition 2.1 (Doxastic grading)**.** Let $\Gamma \neq \emptyset$ be a set of propositional formulae. Let $(G, \succ)$ be a set of grades with a total[1], irreflexive[2] and transitive[3] ordering, from better to worse. Then, a doxastic grading is a function $v : \Gamma \to G$ that grades each propositional formula in terms of how plausible a hypothesis it is.

We assume that G is non-empty, finite, and not necessarily numeric. E.g.:

$$G = \{\mathsf{False}, \mathsf{Bad}, \mathsf{Plausible}, \mathsf{Good}, \mathsf{Very\ Good}, \mathsf{True}\},$$

where $\mathsf{False} \prec \ldots \prec \mathsf{True}$.

In comparison with the categorical belief, doxastic grading allows us to express how strongly propositions are (dis)believed. Doxastic grading is also more expressive than relative belief. With the relative belief, we can say that one proposition is more plausible than the other, but we cannot say anything about how drastic the difference is. However, qualitatively graded beliefs are expressively weaker than the probabilistic ones: with qualitative grades, we cannot numerically express the difference in degrees of (dis)belief.

So given that qualitative grades can express less than probability distributions, why would we not just stick to probabilities instead? Our argument comes from a certain epistemological position: *evidentialism*. Briefly, evidentialism claims that a doxastic attitude to a proposition is justified iff it fits the evidence. Oftentimes, the evidence[4] an agent has is qualitative, not quantitative. The evidence may neither indicate the frequency or propensity of propositions being true, nor allow the agent to numerically measure her degrees of certainty in the propositions. In the spirit of evidentialism, we argue that it is unjustified to ascribe subjective probabilities to the propositions in such cases.

For example, assume that a traveller waits for a train in a country that she has never visited before. She has heard some rumours that the trains in this land tend to be late. A timetable and railway workers reassure her that the train will be on time. In this situation, the traveller has reasons to believe that the train will not be late: both the timetable and the railway workers appear to be more reliable sources of information than the rumours. However, she cannot express her beliefs in the form of a probability distribution. As it is the first time she tries to take this train, the event is unique. All the evidence she has gathered from the rumours, the timetable, and the workers is non-quantitative. She also knows nothing about how often the sources of the testimonies she has received tend to be accurate. So ascribing any precise probability to the event of the train coming on time seems to be unjustified. The traveller still can express her doxastic attitude either as a relative belief (*it is more plausible that the train will be on time than not*) or via the doxastic grading (*the train will very likely be on time, and it is unlikely that the train will be late*).

The key point is: agents should assign higher grades to propositions that are supported by better evidence, and they can do so qualitatively when the evidence does not allow them to do so numerically. Understanding the rationality norms of graded belief, thus, crucially reduces to understanding how rational agents form their doxastic attitudes from non-quantitative, possibly conflicting evidence of varying quality. In the following section, we introduce a formal apparatus we will use to model a rational agent's formation of doxastic attitudes that is based on heterogeneous pieces of evidence.

3 Formal background: topology of evidence

For exposition of our formal framework, we will need to represent an agent forming her beliefs based on the available evidence. We will take our cue from the *topological models of belief*, where pieces of evidence are represented as open sets in a topological space (Van Benthem and Pacuit 2011; Baltag, Bezhanishvili, et al. 2022). With the help of these topological models for epistemic and doxastic logic, a variety of categorical epistemic attitudes, such as fallible knowledge, knowledge with certainty, etc., were successfully formalised. Given our hypothesis that doxastic grading is grounded in having qualitative evidence, the topological semantics is a good candidate for a foundation to build on. In this section, we introduce the required topological and doxastic-logical background for our needs.

3.1 Preliminaries: basic topology

Definition 3.1 (Topological space)**.** Topological space is a pair (X, τ), where X is a non-empty set, and $\tau \subseteq 2^X$ is a collection of X's subsets, such that:

[1] For each $\alpha, \beta \in G$, either $\alpha \succ \beta$, or $\beta \succ \alpha$, or $\alpha = \beta$.

[2] For each $\alpha \in G$, $\alpha \not\succ \alpha$.

[3] For each $\alpha, \beta, \gamma \in G$, if $\alpha \succ \beta$ and $\beta \succ \gamma$, then $\alpha \succ \gamma$.

[4] We stay agnostic with respect to the nature of the evidence. namely, we do not make any assumptions as to whether the basic pieces of evidence the agent has are propositions, sensory inputs, testimonies, etc. We take a piece of evidence to be anything that provides an agent with reasons to believe that some propositions are true.

1. $\emptyset, X \in \tau$;
2. If $U_1, U_2 \in \tau$, then $U_1 \cap U_2 \in \tau$;
3. If $\{U_i\}_{i \in I} \subseteq \tau$ for some arbitrary index-set I, then $\bigcup_{i \in I} U_i \in \tau$

X is called *space*, and τ – *topology on* X. Elements of τ are called *open sets* (or shortly, opens). For every point $x \in X$, U is a neighborhood of x iff $x \in U \in \tau$. A *closed set* is a set $C \subseteq X$ that is the complement of some open. By $\overline{\tau} = \{C \subseteq X \mid \exists U \in \tau : C = X \setminus U\}$ we denote a collection of closed sets of a topological space (X, τ).

Given an arbitrary collection of subsets $\mathcal{B} \subseteq 2^X$, we say that $\mathcal{B}$ is the *subbasis* for τ (or that $\mathcal{B}$ *generates* τ) iff τ is the smallest topology containing elements of $\mathcal{B}$ as its opens. If $\mathcal{B}$ is a subbasis of τ, then all finite intersections of elements of $\mathcal{B}$, i.e., $\{O \subseteq X \mid O = E_1 \cap \ldots \cap E_n$ for some $E_1, \ldots, E_n \in \mathcal{B}\}$, is a *basis* of τ. Moreover, τ may be seen as the collection of arbitrary unions of finite intersections of elements of $\mathcal{B}$, together with X and $\emptyset$.

Intuitively, it is convenient to think of X as a set of points in space, where opens are observable regions of that space. That allows us to topologically capture the notions of closeness or nearness, without necessarily having a quantitative measure defined in the space.

Given any subset $A \subseteq X$, by the *interior* of A, denoted as $Int(A)$, we mean the largest open set that is *contained* in A, i.e., $Int(A) = \bigcup\{U \in \tau \mid U \subseteq A\}$. By the *closure* of A, denoted as $Cl(A)$, we mean the smallest closed set that is *containing* A, i.e., $Cl(A) = \bigcap\{C \in \overline{\tau} \mid A \subseteq C\}$. The interior and closure operators have the following algebraic properties, known as Kuratowski axioms:

$$\begin{array}{ll} Int(X) = X & Cl(\emptyset) = \emptyset \\ Int(A) \subseteq A & A \subseteq Cl(A) \\ Int(A \cap B) = Int(A) \cap Int(B) & Cl(A \cup B) = Cl(A) \cup Cl(B) \\ Int(Int(A)) = Int(A) & Cl(Cl(A)) = Cl(A) \end{array}$$

3.2 Preliminaries: epistemic and doxastic logic

By epistemic and doxastic logic, we mean modal logics, where $\Box$-operator is interpreted as an agent's knowledge or belief. Formally, let $Prop = \{p_1, p_2, \ldots\}$ be a countable set of propositional variables. Then, we recursively define a formal language of epistemic logic $\mathcal{L}_{EL}$ by the following grammar:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \vee \varphi) \mid K\varphi \mid B\varphi$$

where $p \in Prop$. Boolean connectives $(\neg, \vee)$ have their standard meaning. We also use standard acronyms for other Boolean connectives: $\varphi \wedge \psi := \neg(\neg\varphi \vee \neg\psi)$, $\varphi \to \psi := \neg\varphi \vee \psi$, $\varphi \leftrightarrow \psi := (\varphi \to \psi) \wedge (\psi \to \varphi)$. $K\varphi$ reads as *agent knows that* φ. $B\varphi$ reads as *agent believes that* φ. Both operators have their defined duals: $\widehat{K}\varphi := \neg K \neg\varphi$ reads as *the agent considers it possible that* φ; $\widehat{B}\varphi := \neg B \neg\varphi$ – *φ is consistent with agent's beliefs.*

Definition 3.2 (Epistemic plausibility models)**.** An epistemic plausibility model is a tuple of the form $M = (W, \leq, V)$, where $\leq$ is a converse well-founded preorder on W. $\leq$ being a preorder means that it is reflexive and transitive. $\leq$ being converse well-founded means that every non-empty subset of W has a maximal element. That is, for every $\emptyset \neq X \subseteq W$, the set $Max_X = \{w \in X \mid \forall v \in X(v \leq w)\}$ is non-empty. $V : Prop \to 2^W$ is a valuation function, mapping each propositional variable to the set of possible worlds, in which the corresponding proposition is evaluated as true.

Informally, W is the set of possible words, or states how the world might be. We assume that the agent does not know which of the possible worlds is the real one, or, using the common terminology in epistemic logic, all worlds in W are indistinguishable for the agent. For example, I do not know whether it is going to rain in Amsterdam tomorrow. So I do not distinguish a possible world where it is going to rain, and the one where it is not.

$\leq$ relation represents the agent's beliefs about which world is most likely to be the real one: for any possible worlds $w, v \in W$, $w \leq v$ iff the agent believes that v is at least as likely to be the real world as w. Converse well-foundedness of $\leq$ guarantees that there will always be some most plausible worlds, according to the agent's beliefs.

$Prop$ is a set of propositional variables that represent some atomic propositions about the worlds, such as "*the capital of the Netherlands is Maastricht*", "*it is going to rain tomorrow*", "*the password for my Google account has 15 symbols*", "*Knaster-Tarski theorem is false*", etc. Some of these propositions may not be true in the actual world (as in the first one we have given), and some might even be false in

all logically possible worlds (as in the last one we have given). V tells us which propositions hold at which worlds. Then, we say that the agent *knows* that φ iff φ is true in all possible worlds, indistinguishable for her. The agent *believes* that φ iff φ is true in all the most plausible possible worlds. Formally, we can interpret $\mathcal{L}_{EL}$ over the class of plasubility models by the following inductive definition:

Definition 3.3 (Semantic clauses for $\mathcal{L}_{EL}$)**.** Given any plausibility model $M = (W, \leq, V)$ and any possible world $w \in W$, we inductively define $\models$ relation as follows:

$M, w \models p$	iff	$w \in V(p)$
$M, w \models \neg\varphi$	iff	$M, w \not\models \varphi$
$M, w \models \varphi \vee \psi$	iff	$M, w \models \varphi$ or $M, w \models \psi$
$M, w \models K\varphi$	iff	$\forall v \in W : M, v \models \varphi$
$M, w \models B\varphi$	iff	$\forall v \in Max_W : M, v \models \varphi$

where $p \in Prop$. By $M \models \varphi$ we mean that $M, w \models \varphi$ for any $w \in W$. By $\models \varphi$ we mean that $M \models \varphi$ for any plausibility model M. Given a set of formulae $\Gamma \subseteq \mathcal{L}_{EL}$, by $M, w \models \Gamma$ we mean that $M, w \models \varphi$ for all $\varphi \in \Gamma$. Given a set of formulae Γ and a formula φ, by $\Gamma \models \varphi$ we mean that if $M, w \models \Gamma$, then $M, w \models \varphi$ for any model M and any possible world of that model. Consequently, $\emptyset \models \varphi$ means that φ is true in all possible worlds of all models. We will abbreviate $\emptyset \models \varphi$ simply as $\models \varphi$, and read it as "φ *is valid*".

The following formulae, each expressing a rationality norm, are valid:

$$\models K(\varphi \to \psi) \to (K\varphi \to K\psi) \tag{1}$$
$$\models B(\varphi \to \psi) \to (B\varphi \to B\psi) \tag{2}$$
$$\models K(\varphi \wedge \psi) \leftrightarrow (K\varphi \wedge \psi) \tag{3}$$
$$\models B(\varphi \wedge \psi) \leftrightarrow (B\varphi \wedge B\psi) \tag{4}$$
$$\models K\varphi \to \varphi \tag{5}$$
$$\models B\varphi \to \neg B\neg\varphi \tag{6}$$
$$\models K\varphi \to B\varphi \tag{7}$$
$$\models B\varphi \to KB\varphi \tag{8}$$
$$\models \neg B\varphi \to K\neg B\varphi \tag{9}$$

(1) and (2) correspond to the closure of the agent's knowledge and belief under logical consequence: whenever the agent knows/believes that φ entails ψ and she knows/believes that φ, she knows/believes that ψ. (3-4) reflects the norm of aggregation: the agent knows/believes that φ and ψ iff the agent knows/believes that φ and the agent knows/believes that ψ. (5) expresses the factivity of knowledge: whenever the agent knows that φ, φ is true. (6) expresses the consistency of belief: whenever the agent believes φ, she does not believe not-φ. (7) states that knowledge implies belief. (8) and (9) express positive and negative introspection, respectively: if the agent believes/does not believe that φ, then she knows that.

3.3 Topology as semantics for epistemic and doxastic logic

Although the plausibility models provide us with a powerful tool to study the logical properties of a rational agent's beliefs, it takes plausibility as the atomic notion. We abstract away from the reasons why the agent considers some possible worlds more plausible than others. As commonly argued for by epistemologists, the rational agent should believe φ iff she has the justification for φ. Oftentimes, the justification is considered to be evidence-based: so to believe that φ, one should have some evidence to support φ. This motivates the search for a finer-grained semantics for epistemic and doxastic logic. One of the major candidates is the *topological semantics*, initially introduced in (McKinsey and Tarski 1944).

Topological spaces turn out to be an elegant way to model evidence. Namely, let $(X, \mathcal{B}_\tau, \tau)$ be a topological space, where $\mathcal{B}_\tau$ is τ's subbasis. We can perceive X as the set of possible words. Then, $\mathcal{B}_\tau$ may be interpreted as the collection of directly observable evidence that the agent obtains via, e.g., testimony, basic belief, experiment, measurement, etc. Then, τ is the collection of properties that the agent can verify via combining her pieces of evidence. To remind the reader, τ is the collection o arbitrary unions of finite intersections of $\mathcal{B}_\tau$'s elements. Taking finite intersections of $\mathcal{B}_\tau$'s elements corresponds to combining pieces of evidence together. For instance, if I take two imprecise measurements of my body temperature, where the first shows that it is between 35.5 and 37 degrees, and the second shows that it is between 36 and 38 degrees, I can combine these measurements and verify that my body temperature

is between 36 and 37 degrees. Taking arbitrary unions of such combined pieces of evidence corresponds to generalizing over them. For instance, assume I know that the value of some variable x is a natural number without knowing which one, and I wonder whether some statement $\varphi(x)$ about x is true. For every natural number $n \in \mathbb{N}$, I have the basic piece of evidence O_n, which indicates that $x = n$ and $\varphi(n)$ holds. Then, I can support my belief in $\varphi(x)$ by generalizing $\{O_n\}_{n\in\mathbb{N}}$ to $\bigcup_{n\in\mathbb{N}} O_n$ (no matter what the value of x is, there is evidence in favour of $\varphi(x)$ for every possible value).

Yet, basic pieces of evidence may be dismissive and mutually inconsistent. How can we capture the properties of evidence that are sufficient for it to be a justification? Here, we stick to the "coherentist" solution of (Baltag, Bezhanishvili, et al. 2016): the body of evidence is strong enough to justify φ iff it supports φ and is consistent with all other pieces of evidence the agent has.

Definition 3.4 (Topological evidence model)**.** A topological evidence model (or, in short, a topo-e-model) is a tuple $M = (W, \mathcal{E}_0, \tau, V)$, where:

1. $W \neq \emptyset$ is a set of possible worlds;
2. $\mathcal{E}_0 \subseteq 2^W$ is s a family of sets called basic evidence sets (or basic opens), satisfying $W \in \mathcal{E}_0$ and $\emptyset \notin \mathcal{E}_0$;
3. $\tau \subseteq 2^W$ is the topology on W, generated by $\mathcal{E}_0$;
4. $V : Prop \to 2^W$ is a standard evaluation function, mapping every propositional variable to a set of possible worlds, where the proposition is true.

Every open $U \in \tau$ is equivalent to an arbitrary union of finite intersections of $\mathcal{E}_0$'s elements. Informally, every open is an *argument*, based on a body of evidence obtained via combining basic pieces of evidence and/or generalising over them. An open $U \in \tau$ is a *justification* iff for any other non-empty open $U' \in \tau \setminus \{\emptyset\}$, $U \cap U' \neq \emptyset$. Or, equivalently, iff $Cl(U) = W$. Topologically, justifications are *dense* opens of τ. Informally, a justification is an argument that is coherent with all other evidence the agent has.

The topo-e models reflect Lehrer's coherentist idea of what she calls *subjective justification*: one is justified in believing that φ iff φ coheres with one's "system of acceptance" (Lehrer 1999). Consequently, we get the following way to interpret $\mathcal{L}_{EL}$ over topo e-models. Given any topo-e model $M = (W, \mathcal{E}_0, \tau, V)$ and any world $w \in W$, $\models$ relation is inductively defined as follows (where cases for propositional variables, Boolean connectives and K-modality are defined as in Def. 3.3):

$$M, w \models B\varphi \text{ iff } \exists U \in \tau(Cl(U) = W \text{ and } U \subseteq [\![\varphi]\!]_M)$$

where $[\![\varphi]\!]_M = \{v \in W \mid M, v \models \varphi\}$ is the truth-set of φ. Informally, $M, w \models B\varphi$ iff there is a body of evidence that supports φ and coheres with all other evidence the agent has.

Such a semantics still validates (1-9) on all topo-e models, preserving the desirable logical properties, while having finer-grained models.

4 The logic of graded evidential support

We differentiate pieces of evidence in terms of their reliability, depending on the sources from which they come. As we conjecture, these differences ground doxastic grading. In topo-e models, we abstract away from these differences: we do not distinguish between direct observations, testimonies, biases, gut feelings, revelations, experimental data, and formal proofs, treating them on par as just pieces of evidence. This obstacle creates some problems for the coherentist definition of justification we employ. On the one hand, we require the justification to cohere even with very unreliable evidence. On the other hand, it suffices to get a very logically weak piece of evidence, and no matter how unreliable it is, it will fall under our definition of justification.

To clarify the problem that we want to point out, think of a horoscope. It is well known that horoscopes appear to be trustworthy, since the predictions they give are so vague and imprecise that it is rare to experience anything inconsistent with these predictions. According to our definition, horoscopes tend to justify beliefs!

Example 1. Assume an agent named Inne, who forms her beliefs via topo e-models. Inne is concerned about feeling tired all the time. She trusts evidence-based medicine, so she visits a doctor and has a blood test performed. The doctor analyses the results and tells Inne: "You have very low iron levels. You will be prescribed a medication. Unless you are a genetic outlier who responds to it super-quickly, you will keep experiencing fatigue for at least a week before the medication starts working. Simple stimulants like caffeine will not help". The next day, Inne goes to a supermarket and buys an energy drink. The

drink's advertisement promises an instant energy boost throughout the day. While enjoying her drink, Inne reads a horoscope, which vaguely says that she will feel neither her best nor her worst that day. What is Inne justified in believing about her well-being that day?

Inne has received three basic pieces of evidence: the doctor's analysis of her blood test results, the horoscope's prediction, and the advertisement's promise about the energy drink's effect. Let us denote them as O_1, O_2, and O_3, respectively. O_1 and O_3 are inconsistent: the doctor predicted Inne feeling fatigue that energy drinks cannot alleviate, while the advertisement promises that Inne will instantly feel full of energy no matter what. O_2, being so vague and imprecise, is coherent with both O_1 and O_3: one may feel fatigued or energetic, while still feeling neither one's best nor one's worst. Therefore, unlike O_1 and O_3, O_2 can serve as a justification, according to our definition.

Formally, let $Prop = \{worst, best, fatigue\}$ be a set of propositional variables, standing for "*Inne feels her worst*", *Inne feels her best*", "*Inne feels fatigue*", respectively. let $M = (W, \mathcal{E}_0, \tau, V)$ be a topo-e model, where $W = \{w_1, \ldots, w_4\}$ is the set of possible worlds, where w_1 is the state, where Inne feels her absolute worst, having all negative conditions possible, including fatigue; w_2 – Inne is fatigued, but not feeling her worst; w_3 – Inne is energized, but still does not feel her best, probably having other issues; w_4 – Inne feels her best, having no issues whatsoever. $\mathcal{E}_0 = \{W, O_1, O_2, O_3\}$ is the set of Inne's basic evidence pieces, where $O_1 = \{w_1, w_2\}$ (Inne feels fatigued, as the doctor predicted), $O_2 = \{w_2, w_3\}$ (Inne feels neither the best nor the worst, as the horoscope predicted), $O_3 = \{w_3, w_4\}$ (Inne feels energised, as the advertisement promised). τ is the topology generated by $\mathcal{E}_0$, that is:

$$\begin{aligned}&\tau = \\ &\{O_1, O_2, O_3, \\ &O_1 \cap O_2, O_2 \cap O_3, \\ &O_1 \cup O_2, O_2 \cup O_3, \\ &\emptyset, W\}\end{aligned}$$

V is defined accordingly: $V(worst) = \{w_1\}$, $V(best) = \{w_4\}$, $V(fatigue) = \{w_1, w_2\}$.

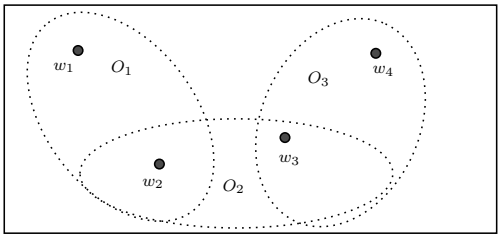


Figure 1: Topo-e model $M = (\{w_1, \ldots, w_4\}, \{O_1, O_2, O_3\}, \tau, V)$. Propositions' evaluations are omitted.

It is not hard to see that only four open sets of τ are dense: O_2, $O_1 \cup O_2$, $O_2 \cup O_3$, and trivially, W. So every justification in τ contains O_2. Let the real world be w_2: Inne is going to feel fatigued, but not at her worst. Then, we can check that the following holds: $M, w_2 \models B\neg(worst \vee best) \wedge \neg B fatigue$. Inne believes what the horoscope tells her, since there exists an open set, O_2, which is dense and which supports $\neg(min \vee \max)$:

$$\|\neg(min \vee max)\|_M = W \setminus (V(min) \cup V(max)) = \{w_2, w_3\} \supseteq O_2 \in \{U \in \tau \mid Cl(U) = W\}$$

On the other hand, Inne cannot justify what the doctor tells her. Two opens support $fatigue$: O_1 and $O_1 \cap O_2$. Both are inconsistent with O_3, hence, not dense.

Intuitively, Inne may still believe she will be fatigued, despite the advertisement's evidence to the contrary. The evidence supporting that she will be fatigued is way more reliable than the one undermining it, so she might ignore what the advertisement says in that case. As for the horoscope prediction, this evidence is also highly unreliable. Its only epistemic merit is its logical weakness, caused by vagueness.

Although the evidence pieces O_2 and O_3 are highly unreliable, just ignoring them altogether may not be the best idea. Given that Inne's epistemic life is still affected by advertisements and horoscopes, she might still use this evidence if she reasons about other propositions or receives some new evidence afterwards. For example, assume Inne is asked to bet on whether she has some disease, undiscovered by modern medicine, that will cause her to feel the worst that day. The most reliable evidence she has, the doctor's analysis, is irrelevant to this proposition, and she has nothing better to rely upon than the horoscope. Or, assume that Inne gets the infallible confirmation that she actually is the outlier who

is super-responsive to the medication the doctor prescribed her, so the iron deficit does not affect her anymore. Then, she may still rely on the bias she gets from the advertisement to assess whether she will be fatigued or not.

To account for such cases, we propose to equip the topo-e models with an explicit representation of the *subjective reliability* of evidence pieces. By subjective reliability, we understand agents' assessment of how likely pieces of evidence are to indicate truth. Subjective reliability should not be confused with truth-tracking, propensity, accuracy scoring, or the success history of mental processes by which the agent obtained her evidence pieces, etc. For example, an agent may subjectively assess her gut feeling as highly reliable, although we may demonstrate that her gut feelings do not track the truth and tend to be false, judging by past experiences.

Assume $(G, \succ)$ to be a finite, linearly and strictly ordered set of qualitative grades the agent ascribes to her evidence pieces, where $\succ$ is a reliability order. That is, if, for any $\alpha, \beta \in G$, $\alpha \succ \beta$ iff α is strictly more reliable than β; $\alpha \succeq \beta$ iff either $\alpha \succ \beta$ or $\alpha = \beta$. We do not presuppose G to be numeric, let alone rational or real: the agent can compare evidence pieces in terms of their reliability, but may not be able to numerically express the degree of that difference.

We postulate two distinguished elements of G. Let $\mathbf{0}, \mathbf{1} \in G$, where $\mathbf{0}$ corresponds to the absolute unreliability, $\mathbf{1}$ – absolute reliability. $\mathbf{0}$ and $\mathbf{1}$ are the minimal and the maximal elements of G, respectively. That is, for any $\alpha \in G$, $\mathbf{0} \preceq \alpha \preceq \mathbf{1}$. We will reserve $\mathbf{0}$ and $\mathbf{1}$ for the absence of evidence and a priori true propositional evidence, respectively.

Definition 4.1 (Graded topo-e model)**.** A graded topo-e model is a tuple $M = (W, \mathcal{E}_0, \tau, g, V)$, where $(W, \mathcal{E}_0, \tau, V)$ is a topo-e model, and $g : \tau \to G$ is a reliability grading function, ascribing any body of evidence with a subjective reliability grade. There are four natural constraints on g:

1. $g(W) = \mathbf{1}$, and for any $U \in \tau$, if $U \neq W$, then $g(U) \neq \mathbf{1}$. A rational agent considers a priori evidence that supports only necessary truths to be absolutely reliable. Only a priori evidence meets this standard of reliability.

2. $g(\emptyset) = \mathbf{0}$, and for any $U \in \tau$, if $U \neq \emptyset$, then $g(U) \neq \mathbf{0}$. A rational agent considers empty evidence that supports everything, including contradictions, to be absolutely unreliable. Only empty evidence meets this standard of reliability.

3. $g(U_1 \cap U_2) \preceq \min\big(\{g(U_1), g(U_2)\}\big)$: the combination of two bodies of evidence is at most as reliable as the least reliable of the two;

4. $g(\bigcup_{i \in I} U_i) \succeq \max(\{g(U_i)\}_{i \in I})$: the generalisation over the set of bodies of evidence is at least as reliable as the most reliable of them.

Having this structure in mind, we recursively define the language of graded evidence and belief $\mathcal{L}$ by the following grammar:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \vee \varphi) \mid B_\alpha \varphi$$

where $p \in Prop$, $\alpha \in G$. Boolean connectives have their standard meaning. $B_\alpha \varphi$ reads as "the best evidence in support of φ the agent has is α-reliable". For $B_{\mathbf{1}}\varphi$, we will use the acronym $\Box\varphi$ – "it is a priori true that φ". By $\mathcal{L}_{CPL} \subseteq \mathcal{L}$ we denote a propositional fragment of $\mathcal{L}$: i.e., if $\varphi \in \mathcal{L}$ and φ contains no modalities, then $\varphi \in \mathcal{L}_{CPL}$.

Definition 4.2. Given a graded topo-e model $M = (W, \mathcal{E}_0, \tau, g, V)$ and a world $w \in W$, $\models$ relation is defined as follows:

$M, w \models p$	iff	$w \in V(p)$
$M, w \models \neg\varphi$	iff	$M, w \not\models \varphi$
$M, w \models (\varphi \vee \psi)$	iff	$M, w \models \varphi$ or $M, w \models \psi$
$M, w \models B_\alpha \varphi$	iff	$\exists U \in \tau (g(U) = \alpha \wedge U \subseteq [\![\varphi]\!]_M) \wedge$
		$\forall U \in \tau (g(U) \succ \alpha \to U \not\subseteq [\![\varphi]\!]_M)$

Informally,$M, w \models B_\alpha \varphi$ iff the agent has evidence to support φ, and the best such evidence she has is α-reliable. Topologically, the semantic clause for $B_\alpha \varphi$ can be redefined as follows:

$$M, w \models B_\alpha \varphi \Leftrightarrow g\big(Int([\![\varphi]\!]_M)\big) = \alpha$$

The logic of graded topo-e models is **L**, as defined in Table 1.

(CPL)	Axioms and inference rules of the classical propositional logic
(K)	$\Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)$
(T)	$\Box\varphi \to \varphi$
(PI)	$B_\alpha\varphi \to \Box B_\alpha\varphi$
(NI)	$\neg B_\alpha\varphi \to \Box\neg B_\alpha\varphi$
(Empty)	$B_{\mathbf{0}}\bot$
(Aggr1)	$(B_\alpha\varphi \wedge B_\beta\psi) \to \bigvee_{\gamma \preceq \min(\{\alpha,\beta\})} B_\gamma(\varphi \wedge \psi)$
(Aggr2)	$(B_\alpha\varphi \wedge B_\beta\psi) \to \bigvee_{\gamma \succeq \max(\{\alpha,\beta\})} B_\gamma(\varphi \vee \psi)$
(Mon)	$\big(B_\alpha\varphi \wedge \Box(\varphi \to \psi)\big) \to \bigvee_{\beta \in \{\gamma \in G \mid g \succeq \alpha\}} B_\beta\psi$
(Un)	$B_\alpha\varphi \to \bigwedge_{\beta \neq \alpha} \neg B_\beta\varphi$
(Compl)	$\bigvee_{\alpha \in G} B_\alpha\varphi$
(RN)	If $\vdash \varphi$, then $\vdash \Box\varphi$
(RE)	If $\vdash \varphi \leftrightarrow \psi$, then $\vdash B_\alpha\varphi \leftrightarrow B_\alpha\psi$
(MP)	If $\vdash \varphi$ and $\vdash \varphi \to \psi$, then $\vdash \psi$

Table 1: Logic **L**

Axioms (K), (T) express that the a priori knowledge is closed under entailment and is factive, respectively. (PI) and (NI) are the axioms of positive and negative introspection: the agent's observed pieces of evidence are accessible to her; she knows what evidence she has or does not have to support each proposition. (Empty) expresses that no non-empty evidence can support a contradiction. (Aggr1) and (Aggr2) axiomatically express the properties of g functions: the combination of two bodies of evidence is at most as reliable as the least reliable of the two, and the generalisation over two bodies of evidence is at least as reliable as the most reliable of the two. (Mon) corresponds to the property that if φ a priori entails ψ, then the evidential support for ψ should be graded at least as high as the one for φ. (Un) and (Compl) say that each piece of evidence receives exactly one grade. (RN) expresses that any logical truth is a priori known, and (RE) – that logically equivalent propositions should have the same evidential support. (MP) is a standard *Modus Ponens* rule of inference.

For any formula $\varphi \in \mathcal{L}$, by $\vdash_{\mathbf{L}} \varphi$ we mean that there exists a derivation of φ in **L**. Formally, a derivation of φ in **L** is a finite sequence of formulae $\varphi_1, \ldots, \varphi_n$, such that:

1. $\varphi_n = \varphi$;
2. Each φ_i is either an axiom of **L**, or is obtained by applying one of the **L**'s inference rules to the previous formulae in a sequence.

For instance, $\vdash_{\mathbf{L}} \Box B_{\mathbf{0}}(p \wedge \neg p)$, as witnessed by the derivation in Table 2.

1.	$\bot \leftrightarrow (p \wedge \neg p)$	(CPL)
2.	$B_{\mathbf{0}}\bot$	(Empty)
3.	$B_{\mathbf{0}}(p \wedge \neg p)$	From (1) and (2) by (RE)
4.	$B_{\mathbf{0}}(p \wedge \neg p) \to \Box B_{\mathbf{0}}(p \wedge \neg p)$	(PI)
5.	$\Box B_{\mathbf{0}}(p \wedge \neg p)$	From (3) and (4) by (MP)

Table 2: A derivation of $\Box B_{\mathbf{0}}(p \wedge \neg p)$ in **L**.

For any set of formulae $\Gamma \subseteq \mathcal{L}$ and a formula $\varphi \in \mathcal{L}$, by $\Gamma \vdash_{\mathbf{L}} \varphi$ we mean that there exists a finite subset $\{\psi_1, \ldots, \psi_n\} \subseteq \Gamma$, such that $\vdash_{\mathbf{L}} (\psi_1 \wedge \ldots \wedge \psi_n) \to \varphi$. When the logical system we work with is clear from the context, we will simply write $\vdash$ instead of $\vdash_{\mathbf{L}}$.

Theorem 4.1. **L** is sound and strongly complete w.r.t. the class of graded topo-e models. In other words, for every set of formulae $\Gamma \subseteq \mathcal{L}$ and any formulae $\varphi \in \mathcal{L}$, the following holds:

$$\Gamma \vdash_{\mathbf{L}} \varphi \Leftrightarrow \Gamma \models \varphi$$

Proof. See the Appendix (Section 8). □

5 From grading reliability to grading propositions

5.1 One-scale threshold model

So far, we have introduced a qualitative scale for grading evidence. An agent might use another scale to grade propositions. E.g., let $(G_{rel}, \succ)$ be a reliability scale for the bodies of evidence, while $(G_{plaus}, \succ)$

is a plausibility scale for propositions. Then, we are confronted with the question of *thresholds*: what standards should the evidential support of the proposition meet to be graded on a plausibility scale in a certain way?

The most straightforward answer is to establish a threshold for every plausibility grade in terms of how good the best evidence in support of the proposition should be to receive this grade. Let $G_{rel} = \{\mathbf{0}, \alpha_1, \ldots, \alpha_n, \mathbf{1}\}$ and $G_{plaus} = \{\beta_1, \ldots, \beta_j\}$ be the reliability, resp., plausibility grading scales, such that $|G_{plaus}| \leq |G_{rel}|$. Then, let $\theta : G_{plaus} \to 2^{G_{rel}}$ be a threshold function. For every $\beta \in G_{plaus}$, $\theta(\beta) \subseteq G_{rel}$ is an interval of grades that satisfy the threshold for β. We assume that θ is injective[5] and $\succ$-monotone.[6]

Definition 5.1 (Doxastic grading)**.** Let $M = (W, \mathcal{E}_0, \tau, g, V)$ be an arbitrary topo-e model, such that $g : \tau \to G_{rel}$. Let G_{plaus} be a plausibility scale, and $\theta : G_{plaus} \to 2^{G_{rel}}$ is a function, such that the set $\{\theta(\beta) \mid \beta \in G_{plaus}\}$ is an interval partition of G_{rel}.[7] For a proposition to be graded with β, the best evidence in favour of this proposition should be α-reliable, for some $\alpha \in \theta(\beta)$. Let $\Gamma \subseteq \mathcal{L}_{CPL}$ be an arbitrary set of propositional formulae. Then, a doxastic grading, yielded by M, is a function $v_M : \Gamma \to G_{plaus}$, defined as follows:

$$v_M(\varphi) = \beta \text{ iff } M \models B_\alpha \varphi \text{ for some } \alpha \in \theta(\beta)$$

In other words, φ is graded with β by v_M iff the reliability grade of the best evidence in support of φ in M lies in the interval $\theta(\beta)$. This threshold semantics yields three natural rationality constraints:

1. If a proposition is a priori true, then it should receive the highest grade on the plausibility scale.
2. If a proposition is a priori false, then it should receive the lowest grade on the plausibility scale.
3. If φ entails ψ, then ψ should be graded at least as high as φ.

Proposition 1. Given any graded topo-e model $M = (W, \mathcal{E}_0, \tau, g, V)$ and a yileded doxastic grading $v_M : \Gamma \to G_{plaus}$, the following holds:

$$\forall \varphi \in \Gamma : \text{ if } M \models \varphi, \text{ then } v_M(\varphi) = \max(G_{plaus}) \tag{10}$$

$$\forall \varphi \in \Gamma : \text{ if } M \models \neg\varphi, \text{ then } v_M(\varphi) = \min(G_{plaus}) \tag{11}$$

$$\forall \varphi, \psi \in \Gamma : \text{ if } M \models \varphi \to \psi, \text{ then } v_M(\varphi) \preceq v_M(\psi) \tag{12}$$

Proof. (10) For the contradiction, assume that $\varphi \in \Gamma$, $M \models \varphi$ and $v_M(\varphi) \neq \max(G_{plaus})$. Since $M \models \varphi$, $[\![\varphi]\!]_M = W$. i.e., $M \models \Box\varphi$. As $v_M(\varphi) \neq \max(G_{plaus})$, $v_M(\varphi) = \beta$ for some $\beta \in G_{plaus}$, such that there exists $\beta' \in G_{plaus}$, such that $\beta' \succ \beta$. Since $M \models \Box\varphi$ and $v_M(\varphi) = \beta$, $\mathbf{1} \in \theta(\beta)$. As θ is $\succ$-monotone, $\forall \alpha \in \theta(\beta), \alpha' \in \theta(\beta') : \alpha' \succ \alpha$. But $\mathbf{1} \in \theta(\beta)$, and $\mathbf{1}$ in the $\prec$-maximal element of G_{rel}. Contradiction. Hence, $v_M(\varphi) = \max(G_{plaus})$.

(11) For the contradiction, assume that $\varphi \in \Gamma$, $M \models \neg\varphi$ and $v_M(\varphi) \neq \min(G_{plaus})$. That is, $v_M(\varphi) = \beta$ for some β, such that there exists $\beta' \in G_{plaus}$, such that $\beta' \prec \beta$. As $M \models \neg\varphi$, $[\![\varphi]\!]_M = \emptyset$. Hence, $M \models B_{\mathbf{0}}\varphi$. Then, $\mathbf{0} \in \theta(\beta)$. As θ is monotone, $\forall \alpha \in \theta(\beta), \alpha' \in \theta(\beta') : \alpha' \prec \alpha$. But $\mathbf{0}$ is the $\prec$-minimal element of G_{rel} and $\mathbf{0} \in \theta(\beta)$. Contradiction. Hence, $v_M(\varphi) = \min(G_{plaus})$.

(12) Assume that $M \models \varphi \to \psi$ for some $\varphi, \psi \in \Gamma$. Then, $[\![\varphi]\!]_M \subseteq [\![\psi]\!]_M$. In other words, $M \models \Box(\varphi \to \psi)$. Let $v_M(\varphi) = \beta$. Then, $M \models \bigvee_{\alpha \in \theta(\beta)} B_\alpha \varphi$. W.l.o.g., let $M \models B_\gamma \varphi$ for some $\gamma \in \theta(\beta)$. For the sake of contradiction, assume that $v_M(\psi) \prec v_M(\varphi)$. Then, $M \models B_\alpha \psi$ for some $\alpha \in \bigcup\{\theta(\beta') \mid \beta' \prec \beta\}$. As θ is $\prec$-monotone and $M \models B_\gamma \varphi$ for some $\gamma \in \theta(\beta)$, $\alpha \prec \gamma$. But from the (Mon) axiom of $\mathbf{L}$, if $M \models B_\gamma \varphi \wedge \Box(\varphi \to \psi)$, then $M \models \bigvee_{\alpha \preceq \gamma} B_\alpha \psi$. Contradiction. Hence, $v_M(\psi) \succeq v_M(\varphi)$. □

Notably, the unrestricted domain assumption – any doxastic grading $v : \Gamma \to G_{plaus}$ should be allowed – still stands, if we assume that Γ consists of logically contingent and mutually exclusive formulae.

Proposition 2. Let $\Gamma \subseteq \mathcal{L}_{CPL}$ be a set of propositions, such that:

1. For any $\varphi \in \Gamma$, $\nvdash \varphi$ and $\nvdash \neg\varphi$;
2. For any $\varphi, \psi \in \Gamma$, if $\varphi \neq \psi$, then $\vdash \neg(\varphi \wedge \psi)$.

[5] For any $\alpha, \beta \in G_{plaus}$, if $\alpha \neq \beta$, then $\theta(\alpha) \neq \theta(\beta)$: distinct grades have distinct thresholds.

[6] For any $\beta_1, \beta_2 \in G_{plaus}$, if $\beta_1 \preceq \beta_2$, then for any $\alpha_1 \in \theta(\beta_1)$, $\alpha_2 \in \theta(\beta_2)$: $\alpha_1 \preceq \alpha_2$: the higher the grade, the higher the threshold.

[7] $\{\theta(\beta) \mid \beta \in G_{plaus}\}$ is an interval partition of G_{rel} iff 1) for any $\beta, \beta' \in G_{plaus}$: $\theta(\beta) \cap \theta(\beta') = \emptyset$; 2) $\bigcup_{\beta \in G_{plaus}} \theta(\beta) = G_{rel}$; 3) For any $\beta \in G_{plaus}$ and for any $\alpha_1, \alpha_2, \alpha_3 \in G_{rel}$: if $\alpha_1, \alpha_3 \in \theta(\beta)$ and $\alpha_1 \preceq \alpha_2 \preceq \alpha_3$, then $\alpha_3 \in \theta(\beta)$.

Then, for every doxastic grading $v : \Gamma \to G_{plaus}$, there exists a reliability scale G_{rel}, a threshold function $\theta : G_{plaus} \to 2^{G_{rel}}$ and a graded topo-e model $M = (W, \mathcal{E}_0, \tau, g, V)$, such that $v(\varphi) = v_M(\varphi)$ for any $\varphi \in \Gamma$.

Proof. Let $(G_{rel}, \preceq)$ be a reliability scale, such that $G_{rel} = G_{plaus} \cup \{\mathbf{0}, \mathbf{1}\}$, and the order of plausibility grades is preserved. Let $\theta : G_{plaus} \to 2^{G_{rel}}$ be a threshold function, such that $\theta(\min(G_{plaus})) = \{\mathbf{0}, \min(G_{plaus})\}$, $\theta(\max(G_{plaus})) = \{\max(G_{plaus}), \mathbf{1}\}$, and for any $\beta \in G_{plaus}$, such that $\min(G_{plaus}) \prec \beta \prec \max(G_{plaus})$: $\theta(\beta) = \{\beta\}$.

Since the propositional formulae in Γ are consistent and mutually exclusive, by the soundness and completeness of classical propositional logic, there exists a set of possible worlds W and a valuation function V, such that $[\![\varphi]\!] \neq \emptyset$ for any $\varphi \in \Gamma$, and for any $\varphi, \psi \in \Gamma$, if $\varphi \neq \psi$, then $[\![\varphi]\!] \cap [\![\psi]\!] = \emptyset$.

Let $\mathcal{E}_0 = \{[\![\varphi]\!] \subseteq W \mid \varphi \in \Gamma, v(\varphi) \neq \mathbf{0}\} \cup \{W\}$, and let τ be a topology generated by $\mathcal{E}_0$.

Next, we show that there exists a function $g : \tau \to G_{rel}$, such that:

$$g([\![\varphi]\!]) = v(\varphi) \text{ for any } \varphi \in \Gamma, \tag{13}$$

and g satisfies the contraints (1-4) of Definition 4.1. For instance, let g be defined as follows:

$$g(U) = \begin{cases} \mathbf{1} & \text{if } U = W \\ \mathbf{0} & \text{if } U = \emptyset \\ \max\left(\{v(\varphi) \mid \varphi \in \Gamma, [\![\varphi]\!] \subseteq U\}\right) & \text{otherwise} \end{cases}$$

Clearly, $g([\![\varphi]\!]) = v(\varphi)$ for any $\varphi \in \Gamma$. g satisfies the constraints (1-2) of Definition 4.1, directly by the definition of g. Clearly, g is monotonic: if $U \subseteq U'$, then $g(U) \preceq g(U')$. Therefore, the constraints (3-4) of Definition 4.1 are satisfied.

It is straightforward to see that a doxastic grading $v_M : \Gamma \to G_{plaus}$, induced by the model $M = (W, \mathcal{E}_0, \tau, g, V)$, is equal to the initial doxastic grading $v : \Gamma \to G_{plaus}$, so $v(\varphi) = v_M(\varphi)$ for every $\varphi \in \Gamma$. □

To conclude, the simple threshold semantics for the doxastic grading yields some very natural rationality constraints that already undermine the unrestricted domain assumption in the epistemic case. The unrestricted domain still holds, in the general case, if we restrict our attention to cases where the graded propositions are neither tautologies nor contradictions, and are mutually exclusive. Of course, if we go into further details, specifying what reliability grades and thresholds we have in mind, some new constraints may follow. For instance, if we want to reserve a special grade for a priori truths, no rational agent should ascribe this grade to two distinct propositions, unless they are a priori equivalent. If the reliability scale we have in mind is probabilistic, it might be irrational to grade both φ and $\neg\varphi$ with the same grade β, if $\theta(\beta) > 0.5$.

5.2 Two-scale threshold model

The simple threshold semantics for doxastic grading has a downside. When forming the doxastic attitude towards a proposition, the agent would take into account only the reliability of the evidence she has *in support* of this proposition, ignoring the evidence *against it*.

For example, assume that you ask two equally trustworthy friends about today's weather. The first friend tells you that they have seen today's forecast, and the forecast predicted heavy rains all day long. The second friend tells you that they have seen the very same forecast, but, according to them, it was predicted that there will be no rain. Being confused, you ask the third friend, who is more trustworthy, since they attentively watch forecasts every day. They say that they do not remember what exactly was forecasted, but rain was definitely mentioned: it might have been just showers in the evening, but it could have been heavy rains all day long. In this situation, you have equally reliable evidence in support of all day long heavy rains and no rains at all: both hypotheses are supported by a testimony of a trustworthy friend. The only difference is that there is a more reliable piece of evidence – the third friend's testimony – which is consistent with the first friend's story and contradicts what the second friend has told you. Clearly, the hypothesis of heavy rains is better than the one of no rain. If we implement the simple threshold semantics, both hypotheses will receive the same grade.

To account for both pro- and contra-evidence, we can implement a two-coordinate scale: when grading every proposition, we look into the best evidence in favour of and against it:

$$\mathbf{B}_{(\alpha,\beta)}\varphi := B_\alpha\varphi \wedge B_\beta\neg\varphi$$

Consequently, we can define the order to compare two-coordinate grades:

$$(\alpha, \beta) \succeq (\alpha', \beta') \text{ iff } \alpha \succeq \alpha' \text{ and } \beta \preceq \beta'$$

Note that $\succeq \subseteq (G \times G)$ is a partial order. If $\alpha \succ \alpha'$ and $\beta \succ \beta'$, then (α, β) and (α', β') are incomparable. If we have two propositions φ and ψ, such that φ has better evidence in favour of it than ψ, but also the evidence against φ is better than the evidence against ψ, it is unclear which of the two propositions is more plausible.

Idea: To obtain a linear order on the two-coordinate scale, we might either prioritise the evidence in favour of or against each proposition, defining the order $\succ \subseteq G_{rel} \times G_{rel}$ lexicographically. I.e., $(\alpha, \beta) \succ (\alpha', \beta')$ iff either $\alpha \succ \alpha'$ or $\alpha = \alpha'$, $\beta' \succ \beta$. Alternatively, $(\alpha, \beta) \succ (\alpha', \beta')$ iff either $\beta' \succ \beta$ or $\beta = \beta'$, $\alpha \succ \alpha'$. In the first case, we postulate that it is more important for a good hypothesis to have good evidence in favour of it rather than having no good evidence against it. In the second case, we postulate the contrary.

To do: explore whether some additional rationality norms emerge when defining thresholds $\theta : (G_{rel} \times G_{rel}) \to G_{plaus}$ on such a two-coordinate scale.

6 Binarisation of doxastic grading

Disclaimer: this section is very drafty!

As we have mentioned, we routinely present agents' beliefs in the form of either probability distributions or the set of believed formulae. The long-standing problem is how these two forms of belief relate to one another, and whether it is possible to reduce the latter to the former. The most intuitive solution is *a threshold binarisation*: an agent believes that φ iff she ascribes the high enough probability to it. The well-known negative result shows that it is not always possible. There might be a rational agent whose probabilistic beliefs are binarised to irrational categorical beliefs (Kyburg 1961). The impossibility is demonstrated via the *lottery paradox*. Assume that we set the threshold as $\vartheta \in (0.5, 1)$. That is, the agent believes that φ if and only if she ascribes to φ a subjective probability of at least ϑ. Then, assume there is a lottery with n-many tickets, where n is a natural number, which is large enough to satisfy $\frac{n-1}{n} > \vartheta$. As the agent knows, the lottery is fair: each ticket is winning with a probability of $\frac{1}{n}$, and only one of them is winning. Let $Prop = \{p_1, \ldots, p_n\}$ be a set of propositions, where p_i stands for "*the ith ticket is a losing one*". Then, the agent believes that each ticket is a losing one with a probability of $\frac{n-1}{n}$. As $\frac{n-1}{n} > \vartheta$, the agent believes that each ticket in particular is a losing one: $Bp_1 \wedge \ldots \wedge Bp_n$. At the very same time, the probability of all tickets being the losing ones is 0, since the agent knows that the lottery is fair. Hence, she does not believe that all of them are losing: $\neg B(p_1 \wedge \ldots \wedge p_n)$. Then, we have the following:

$$Bp_1 \wedge \ldots \wedge Bp_n \wedge \neg B(p_1 \wedge \ldots \wedge Bp_n),$$

which directly contradicts the rationality norm of belief aggregation (4).

But what about the doxastic grading? Can we binarise the qualitatively graded beliefs via some form of threshold translations? Sadly, the very same lottery paradox can be reformulated in qualitative terms, thereby showing that the threshold translation is not always possible. Namely, observe that the following formula is **L**-consistent (hence, satisfiable on some graded topo-e model), as long as $\alpha \neq \mathbf{1}$ and $\alpha \neq \mathbf{0}$:

$$(B_\alpha \varphi \wedge B_\alpha \psi) \wedge \neg B_\alpha (\varphi \wedge \psi)$$

In other words, from the fact that I have α-reliable evidence in favour of both φ and ψ, it does not follow that I have α-reliable evidence in favour of $\varphi \wedge \psi$: the evidence for $\varphi \wedge \psi$ may be strictly weaker than α.

This reflects our intuition as well. I might have highly reliable evidence in favour of φ and ψ, while all the evidence I have in favour of $\varphi \wedge \psi$ is significantly less reliable. For instance, assume an agent regards a testimony that has at least 0.5 chance of being true as reliable enough. The agent has two friends, each of whom is known to tell the truth half of the time. The friends' testimonies are independent. One friend tells the agent that it will not rain tomorrow, while the other says that it will not be windy either. She has no further evidence about tomorrow's weather. So the agent has reliable enough evidence to support that it will not rain tomorrow, and that it will not be windy. What about the evidence to support that it will be neither rainy nor windy? Two independent testimonies, each having a 50% chance of being true, have only a 25% chance of being simultaneously true. As nothing else supports this proposition, the agent does not have reliable enough evidence to support the claim that tomorrow will be neither rainy nor windy.

Formally, assume we have any pair of plausibility and reliability scales $(G_{plaus}, \succ)$, $(G_{rel}, \succ)$. Let $\theta : G_{plaus} \to 2^{G_{rel}}$ be a threshold function. Further, let $\vartheta \in G_{plaus}$ be a grade such that the agent believes that φ iff $v(\varphi) \succeq \vartheta$. Then, unless $\theta(\vartheta) = \{\mathbf{1}\}$ or $\theta(\vartheta) = \{\mathbf{0}\}$, there exists a topo-e model, such that $v_M(p) = \vartheta$, $v_M(q) = \vartheta$, while $v_M(p \wedge q) \prec \vartheta$. To show this, assume that $\theta(\vartheta) = \{\alpha_1, \ldots, \alpha_n\}$ for any $\alpha_1, \ldots, \alpha_n \in G_{rel}$, such that $\alpha_1 \prec \ldots \prec \alpha_n$. Then, the following formula is **L**-consistent:

$$B_{\alpha_1}p \wedge B_{\alpha_1}q \wedge \bigvee_{\beta \prec \alpha_1} B_\beta(p \wedge q)$$

As this formula is **L**-consistent and **L** is sound and complete w.r.t. graded topo-e models, there exists a model $M = (W, \mathcal{E}_0, \tau, V)$ and a world $w \in W$, such that $M, w \models B_{\alpha_1}p \wedge B_{\alpha_1}q \wedge \bigvee_{\beta \prec \alpha_1} B_\beta(p \wedge q)$. As $\theta(\vartheta) = \{\alpha_1, \ldots, \alpha_n\}$, $v_M(p) = \vartheta$, $v_M(q) = \vartheta$, $v_M(p \wedge q) \prec \vartheta$. Then, the agent believes p and the agent believes q, but the agent does not believe $p \wedge q$, which contradicts the rationality norm of belief aggregation (4).

Another intuitively appealing approach to the binarisation of qualitatively graded belief might be as follows. The agent should believe that φ iff the evidence in favour of φ is more reliable than the evidence against it:

$$B\varphi \leftrightarrow \bigvee_{\alpha \in G_{rel}} (B_\alpha \varphi \wedge \bigvee_{\beta \prec \alpha} B_\beta \neg\varphi)$$

Unfortunately, the very same kind of paradoxes can be used against it. Recall the axioms (Aggr1) and (Aggr2):

$$\models (B_\alpha \varphi \wedge B_\beta \psi) \rightarrow \bigvee_{\gamma \succeq \max(\{\alpha, \beta\})} B_\gamma(\varphi \vee \psi)$$

$$\models (B_\alpha \varphi \wedge B_\beta \psi) \rightarrow \bigvee_{\gamma \preceq \min(\{\alpha, \beta\})} B_\gamma(\varphi \wedge \psi)$$

That is, the evidential support of $\varphi \vee \psi$ is at least as reliable as the support of φ and ψ, and the evidential support of $\varphi \wedge \psi$ is at most as reliable as the support of φ and ψ.

So, assume that the best evidence the agent has in favour of φ and ψ is strictly more reliable than the evidence she has against φ and ψ:

$$B_\alpha \varphi \wedge B_{\alpha'} \psi \wedge B_\beta \neg\varphi \wedge B_{\beta'} \neg\psi, \text{ where } \alpha \succ \beta, \alpha' \succ \beta'$$

Then, the evidence in favour of $\varphi \wedge \psi$ may be strictly less reliable than $\min(\{\alpha, \alpha'\})$, while the evidence in favour of $\neg\varphi \vee \neg\psi$ may be strictly more reliable than $\max\{\beta, \beta'\}$. Let $\gamma, \delta \in G_{plaus}$ be the grades, such that $\gamma \succ \max\left(\{\alpha, \alpha', \beta, \beta'\}\right)$, $\delta \prec \min\left(\{\alpha, \alpha', \beta, \beta'\}\right)$ Then, the following formula is **L**-consistent:

$$B_\alpha \varphi \wedge B_{\alpha'} \psi \wedge B_\beta \neg\varphi \wedge B_{\beta'} \neg\psi \wedge B_\delta(\varphi \wedge \psi) \wedge B_\gamma(\neg\varphi \vee \neg\psi)$$

As $\models (\neg\varphi \vee \neg\psi) \leftrightarrow \neg(\varphi \wedge \psi)$, and the equivalent formulae are graded equally, the following is consistent as well:

$$B_\alpha \varphi \wedge B'_\alpha \psi \wedge B_\beta \neg\varphi \wedge B_{\beta'} \neg\psi \wedge B_\delta(\varphi \wedge \psi) \wedge B_\gamma \neg(\varphi \wedge \psi)$$

So, $B_\delta(\varphi \wedge \psi) \wedge B_\gamma \neg(\varphi \wedge \psi)$, while $\gamma \succ \delta$. Therefore, $\varphi \wedge \psi$ is not believed, while φ and ψ are believed, which contradicts (4).

To conclude, the most obvious approaches to the binarisation of qualitatively graded beliefs face the very same paradoxes as the threshold binarisation of probabilistic belief does. To alleviate this, we might look into more refined versions of the binarisation, as, e.g., the stability theory of belief (Leitgeb 2014).

7 Questions to discuss

I will write the conclusion when the paper is finished :) Instead, here's the list of questions about this draft I would like to discuss:

1. Here, I propose the evidentialist semantics for qualitatively graded beliefs. Of course, it is not the only possible one. What other semantics may make sense? E.g., should we rather interpret qualitative plausibility grades as thresholds on subjective probabilities? For instance, I find it highly likely that φ iff $Pr(\varphi) \in [0.8, 0.9)$.

2. Are there any other ways to move from reliability grades to the plausibility ones? In terms of the two-scale threshold model, can't we somehow aggregate the quality of the evidence in favour of and against φ to get a one-dimensional grade? If so, how should we aggregate them? The solutions from voting theory might be useful.

3. Binarisation of qualitatively graded beliefs. Are there any other ways to do so you can think of I should try?

8 Appendix

Theorem 8.1. **L** is sound and strongly complete w.r.t. the class of graded topo-e models. In other words, for every set of formulae $\Gamma \subseteq \mathcal{L}$ and any formulae $\varphi \in \mathcal{L}$, the following holds:

$$\Gamma \vdash_{\mathbf{L}} \varphi \Leftrightarrow \Gamma \models \varphi$$

First, we define a broader class of pseudo-models that we denote as **P**.

Definition 8.1. A pseudo-model is a tuple $M = (W, \mathcal{E}^0, \tau, g, V)$, where $(W, \mathcal{E}^0, \tau, V)$ is a topo-e model, and $g : \tau \to G$ is a grading function, such that:

1. $g(W) = \mathbf{1}$, and for any $U \in \tau$, if $g(U) = \mathbf{1}$, then $U = W$;
2. $g(\emptyset) = \mathbf{0}$. Unlike in graded topo-e models, other opens may be graded with $\mathbf{0}$ as well;
3. $g(U_1 \cap U_2) \preceq \min(g(U_1), g(U_2))$;
4. $g(\bigcup_{i \in I} U_i) \succeq \max(\{g(U_i)\}_{i \in I})$.

Note that pseudo-models differ from graded topo-e models only in one property of grading functions concerning $\emptyset$. While in graded topo-e models, only $\emptyset$ is graded with $\mathbf{0}$, in pseudo-models, other opens may be graded with $\mathbf{0}$ as well.

First, we prove that **L** is strongly sound and complete w.r.t. **P**. Then, we show that $\mathcal{L}$ does not distinguish between graded topo-e models and pseudo-models. I.e., $\Gamma \models_{\mathbf{P}} \varphi$ iff $\Gamma \models \varphi$.

Definition 8.2. A set of formulae $\Gamma \subseteq \mathcal{L}$ is a maximal consistent set (MCS) iff $\Gamma \nvdash_{\mathbf{L}} \bot$ and for every $\psi \in \mathcal{L} \setminus \Gamma$, $\Gamma \cup \{\psi\} \vdash_{\mathbf{L}} \bot$. Given two MCSs Γ, Δ, we say that Γ and Δ are $\Box$-equivalent iff $\{\varphi \in \mathcal{L} \mid \Box\varphi \in \Gamma\} = \{\varphi \in \mathcal{L} \mid \Box\varphi \in \Delta\}$. By $[\Gamma]_\Box$ we denote the set of all MCS that are $\Box$-equivalent to Γ.

Definition 8.3 (Canonical model)**.** For any MCS Γ, we define a canonical model M^c as follows:

$$M^c = \{W^c, \mathcal{E}^c_0, \tau^c, g^c, V^c\}, \text{ where:}$$

1. $W^c = [\Gamma]_\Box$;
2. $\mathcal{E}^c_0 = \{\widehat{\varphi} \mid \exists \alpha \in G : B_\alpha\varphi \in \Gamma\} \setminus \{\emptyset\}$, where $\widehat{\varphi} = \{\Delta \in W^c \mid \varphi \in \Delta\}$;
3. τ^c is a topology generated by $\mathcal{E}^c_0$;
4. $g^c : \tau^c \to G$ is a grading function that is defined as follows:

$$g^c(U) = \begin{cases} \mathbf{0} & \text{if } U = \emptyset \\ \alpha \in G, \text{ s.t. } B_\alpha\varphi \in \Gamma & \text{if } U = \widehat{\varphi} \text{ for some } \widehat{\varphi} \in \mathcal{E}^c_0 \\ \max(\{\alpha \in G \mid B_\alpha\varphi \in \Gamma, \widehat{\varphi} \subseteq U\}) & \text{otherwise} \end{cases}$$

5. $V^c(p) = \widehat{p}$ for every $p \in Prop$.

Lemma 8.2. $g^c : \mathcal{E}^c_0 \to G$ is a well-defined function.

Proof. We need to show that for any $U_1, U_2 \in \tau^c$, if $U_1 = U_2$, then $g^c(U_1) = g^c(U_2)$. We have two cases: either $U_1, U_2 \in \mathcal{E}^c_0$ or $U_1, U_2 \in \tau^c \setminus \mathcal{E}^c_0$ (since $U_1 = U_2$, it is impossible that $U_1 \in \mathcal{E}^c_0$, but $U_2 \notin \mathcal{E}^c_0$, and vice versa).

Case 1: $U_1, U_2 \in \mathcal{E}^c_0$. Then, $U_1 = \widehat{\varphi}$, $U_2 = \widehat{\psi}$ for some $B_\alpha\varphi, B_\beta\psi \in \Gamma$. As $U_1 = U_2$, $\widehat{\varphi} = \widehat{\psi}$, so $\Box(\varphi \leftrightarrow \psi) \in \Gamma$. Then, by (RE) rule, $B_\alpha\varphi \leftrightarrow B_\alpha\psi$. Therefore, $B_\alpha\varphi, B_\alpha\psi \in \Gamma$, $B_\beta\varphi, B_\beta\psi \in \Gamma$. By the axiom (Un) of **L**, if $\alpha \neq \beta$, then $\{B_\alpha\varphi, B_\beta\varphi\} \vdash_{\mathbf{L}} \bot$. As Γ is consistent, $\Gamma \nvdash \bot$. Hence, $\alpha = \beta$, and $g^c(U_1) = g^c(U_2)$.

Case 2: $U_1, U_2 \notin \mathcal{E}^c_0$. Then:

$$\begin{aligned} & g^c(U_1) \\ & = \max(\{\alpha \in G \mid B_\alpha\varphi \in \Gamma, \widehat{\varphi} \subseteq U_1\}) \\ & = \max(\{\alpha \in G \mid B_\alpha\varphi \in \Gamma, \widehat{\varphi} \subseteq U_2\}) \\ & = g^c(U_2) \end{aligned}$$

So in both cases, g^c is well-defined. □

Lemma 8.3. τ^c contains $\mathcal{E}_0^c$ together with the arbitrary unions of $\mathcal{E}_0^c$'s elements and the empty set. Or, formally:

$$\tau^c = \mathcal{E}_0^c \cup \{\bigcup_{i\in I} U_i \mid \{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c\} \cup \{\emptyset\}$$

Proof. Since τ^c is a topology generated from $\mathcal{E}_0^c$,τ^c is the minimal set such that:

1. $\mathcal{E}_0^c \subseteq \tau^c$;
2. $W^c \in \tau^c$;
3. $\emptyset \in \tau^c$;
4. $U_1, U_2 \in \tau^c \Rightarrow U_1 \cap U_2 \in \tau^c$;
5. $\{U_i\}_{i\in I} \subseteq \tau^c \Rightarrow \bigcup_{i\in I} U_i \in \tau^c$.

We show that $E = \mathcal{E}_0^c \cup \{\bigcup_{i\in I} U_i \mid \{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c\} \cup \{\emptyset\}$ is the minimal set that satisfies these properties. Clearly, $W^c \in \mathcal{E}_0^c \subseteq E$, as $B_{\mathbf{1}}\top \in \Gamma$ and $\widehat{\top} = W^c$. By definition of E, $\emptyset \in E$.

E is closed under arbitrary unions. Take an arbitrary collection of elements $\{U_i\}_{i\in I} \subseteq E$. If $\{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c$, then, by definition, $\bigcup_{i\in I} U_i \in E$. If $\{U_i\}_{i\in I} \not\subseteq \mathcal{E}_0^c$, then $\{U_i\}_{i\in I}$ can be partitioned $\{U_i\}_{i\in I} = \{U_j\}_{j\in J} \cup \{U_k\}_{k\in K}$, where $\{U_j\}_{j\in J} = \{U_i\}_{i\in I} \cap \mathcal{E}_0^c$ and $\{U_k\}_{k\in K} \cap \mathcal{E}_0^c = \emptyset$. By definition of E, since every U_k is not an element of $\mathcal{E}_0^c$, it is an arbitrary union of some elements of $\mathcal{E}_0^c$, i.e., $U_k = \bigcup\{U_l^k\}_{l\in L_k}$ for some $\{U_l^k\}_{l\in L_k} \subseteq \mathcal{E}_0^c$. Therefore:

$$\begin{aligned}
&\bigcup_{i\in I} U_i \\
=&\bigcup_{j\in J} U_j \cup \bigcup_{k\in K} U_k \\
=&\bigcup_{j\in J} U_j \cup \bigcup_{k\in K}\bigcup_{l\in L_k} U_l^k
\end{aligned}$$

which is an arbitrary union of $\mathcal{E}_0^c$'s elements. Hence, be definition of E, $\bigcup_{i\in I} U_i \in E$.

So to prove that E is a topology, it is left for us to show that E is closed under binary intersections. Take any $U_1, U_2 \in E$. If $U_1 \cap U_2 = \emptyset$, then $U_1 \cap U_2 \in E$ by definition. Assume $U_1 \cap U_2 \neq \emptyset$. Then we have four cases:

Case 1: $U_1, U_2 \in \mathcal{E}_0^c$. then, $U_1 = \widehat{\varphi}$, $U_2 = \widehat{\psi}$ for some φ, ψ, such that $B_\alpha\varphi, B_\beta\psi \in \Gamma$ for some $\alpha, \beta \in G$. Note that $B_\alpha\varphi \wedge B_\beta\psi \vdash_{\mathbf{L}} \bigvee_{\gamma\preceq\min(\{\alpha,\beta\})} B_\gamma(\varphi\wedge\psi)$ by (Aggr1). As Γ is a MCS and $\{B_\alpha\varphi, B_\beta\psi\} \subseteq \Gamma$, $\bigvee_{\gamma\preceq\min(\{\alpha,\beta\})} B_\gamma(\varphi\wedge\psi) \in \Gamma$. So there is some grade γ, such that $B_\gamma(\varphi\wedge\psi) \in \Gamma$. Hence, $\widehat{\varphi\wedge\psi} = \widehat{\varphi}\cap\widehat{\psi} = U_1 \cap U_2 \in \mathcal{E}_0^c \subseteq E$.

Case 2: $U_1 \in \mathcal{E}_0^c$, $U_2 = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c$. Then, $U_1 \cap U_2 = \bigcup\{U_1 \cap U_i \mid i \in I, U_1 \cap U_i \neq \emptyset\}$. As $U_1 \in \mathcal{E}_0^c$ and $\forall i \in I : U_i \in \mathcal{E}_0^c$, by the previous case, $\{U_1 \cap U_i \mid i \in I, U_1 \cap U_i \neq \emptyset\} \subseteq \mathcal{E}_0^c$. As E contains all the arbitrary unions of $\mathcal{E}_0^c$'s elements, $U_1 \cap U_2 = \bigcup\{U_1 \cap U_i \mid i \in I, U_1 \cap U_i \neq \emptyset\} \in E$.

Case 3: $U_1 = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c$, $U_2 \in \mathcal{E}_0^c$. Analogously to the previous case.

Case 4: $U_1 = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c$, $U_2 = \bigcup_{j\in J} U_j$ for some $\{U_j\}_{j\in J} \subseteq \mathcal{E}_0^c$. Then, $U_1 \cap U_2 = \bigcup\{U_i \cap U_j \mid i \in I, j \in J, U_i \cap U_j \neq \emptyset\}$. By the case 2, for each $i \in I, j \in J$, if $U_i \cap U_j \neq \emptyset$, then $U_i \cap U_j \in \mathcal{E}_0^c$. Hence, $\bigcup\{U_i \cap U_j \mid i \in I, j \in J, U_i \cap U_j \neq \emptyset\}$ is an arbitrary union of $\mathcal{E}_0^c$'s elements, therefore, $U_1 \cap U_2 = \bigcup\{U_i \cap U_j \mid i \in I, j \in J, U_i \cap U_j \neq \emptyset\} \in E$.

Finally, we need to prove that E is the smallest topology that contains $\mathcal{E}_0^c$. By definition, any topology that contains $\mathcal{E}_0^c$ has the empty set as an open set and is closed under arbitrary unions, so it will contain E. I.e., if τ is a topology and $\mathcal{E}_0^c \subseteq \tau$, then $E \subseteq \tau$. As we have proven E to be a topology that contains $\mathcal{E}_0^c$, it is the minimal such topology. □

Corollary 8.3.1. $g^c : \tau^c \to G$ may be equivalently redefined as follows:

$$g^c(U) = \begin{cases} \alpha \in G, \text{ s.t. } B_\alpha\varphi \in \Gamma & \text{if } U = \widehat{\varphi} \text{ for some } \widehat{\varphi} \in \mathcal{E}_0^c \\ \max(\{\alpha \in G \mid B_\alpha\varphi \in \Gamma, \widehat{\varphi} \subseteq U\}) & \text{if } U = \bigcup_{i\in I} U_i \text{ for some } \{U_i\}_{i\in I} \subseteq \mathcal{E}_0^c, \text{ s.t. } \bigcup_{i\in I} U_i \notin \mathcal{E}_0^c \end{cases}$$

Lemma 8.4. For any $U_1, U_2 \in \tau^c$, if $U_1 \subseteq U_2$, then $g^c(U_1) \preceq g^c(U_2)$.

Proof. By induction on U_1, U_2.

Case 1: $U_1, U_2 \in \mathcal{E}^c_0$. Follows directly from axiom (Mon).

Case 2: $U_1 \in \mathcal{E}^c_0$, $U_2 = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq \mathcal{E}^c_0$. Then, $g^c(U_2) = \max(\{g^c(U') \mid U' \in \mathcal{E}^c_0, U' \subseteq U_2\})$. Given that $U_1 \in \mathcal{E}^c_0$ and $U_1 \subseteq U_2$, $g^c(U_1) \preceq \max(\{g^c(U') \mid U' \in \mathcal{E}^c_0, U' \subseteq U_2\})$, i.e., $g^c(U_1) \preceq g^c(U_2)$.

Case 3: $U_1 = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq \mathcal{E}^c_0$, $U_2 \in \mathcal{E}^c_0$. Then, $U_2 = \widehat{\varphi}$ for some $B_\alpha\varphi \in \Gamma$, and $g^c(U_1) = \max\{g^c(U') \mid U' \subseteq U_1, U' \in \mathcal{E}^c_0\}$. Assume that $U' \in \mathcal{E}^c_0$ is a set, such that $U' \subseteq U_1$ and $g^c(U') \succeq g^c(U'')$ for any $U'' \subseteq U_1$. W.l.o.g., let $g^c(U') = \beta$ for some $\beta \in G$. Then, $g^c(U_1) = \beta$. By def. of $\mathcal{E}^c_0$, $U' = \widehat{\psi}$ for some ψ, such that $B_\beta\psi \in \Gamma$. As $U_2 \in \mathcal{E}^c_0$, $U_2 = \widehat{\varphi}$ for some φ, such that $B_\alpha\varphi \in \Gamma$ for some $\alpha \in G$. Since $U' \subseteq U_1 \subseteq U_2$, $U' \subseteq U_2$, hence, $\widehat{\psi} \subseteq \widehat{\varphi}$, i.e., $\Box(\psi \to \varphi) \in \Gamma$. Then, since $B_\beta\psi, \Box(\psi \to \varphi) \in \Gamma$, by the (Mon) axiom, $\bigvee_{\gamma\succeq\beta} B_\gamma\psi$. As $B_\alpha \in \Gamma$, and by the axiom (Un) α is the unique grade for φ, $\alpha \succeq \beta$, i.e., $g^c(U_1) \succeq g^c(U_2)$.

Case 4: $U_1 = \bigcup_{i\in I} U_i$, $U_2 = \bigcup_{j\in J} U_j$ for some $\{U_i\}_{i\in I}, \{U_j\}_{j\in J} \subseteq \mathcal{E}^c_0$. Then, $g^c(U_1) = \max\{g^c(U') | U' \in \mathcal{E}^c_0, U' \subseteq U_1\}$, $g^c(U_2) = \max\{g^c(U') \mid U' \in \mathcal{E}^c_0, U' \subseteq U_2\}$. As $U_1 \subseteq U_2$, the best piece of evidence from $\mathcal{E}^c_0$ that supports U_1 also supports U_2, therefore, $\max\{g^c(U') \mid U' \in \mathcal{E}^c_0, U' \subseteq U_1\} \preceq \max\{g^c(U') \mid U' \in \mathcal{E}^c_0, U' \subseteq U_2\}$, i.e., $g^c(U_1) \preceq g^c(U_2)$. □

Lemma 8.5 ($M^c \in \mathbf{P}$.)**.** *M^c is a pseudo-model.*

Proof. By definition, W^c is a non-empty set, $\mathcal{E}^c_0 \subseteq 2^{W^c}$ is a collection of its subsets, and τ^c is a topology generated by $\mathcal{E}^c_0$. As we have shown in Lemma 8.2, $g^c : \tau^c \to G$ is a well-defined function.

It is left for us to prove that g^c has the needed properties.

We are to prove that $\forall U \in \tau^c : g^c(U) = \mathbf{1}$ iff $U = W^c$. Right-to-left direction. Assume $U = W^c$. Then, $U = \widehat{\top}$. By (RN), $\Box\top \in \Gamma$, so, by def. of $\mathcal{E}^c_0$, $U = \widehat{\top} \in \mathcal{E}^C_0 \subseteq \tau^c$ and $g^c(U) = \mathbf{1}$. Left-to-right:

$$\begin{aligned}
& g^c(U) = \mathbf{1} \\
\Rightarrow \quad & \exists \Box\varphi \in \Gamma : U = \widehat{\varphi} \\
\Rightarrow \quad & \forall \Delta \in [\Gamma]_\Box : \Box\varphi \in \Delta \\
\Rightarrow \quad & \forall \Delta \in [\Gamma]_\Box : \varphi \in \Delta \\
\Rightarrow \quad & W^c = \widehat{\varphi} \\
\Rightarrow \quad & W^c = U
\end{aligned}$$

Note that $g^c(\emptyset) = \mathbf{0}$, directly by the definition of g^c. So it is left for us to show that $g(U_1 \cap U_2) \preceq \min(g^c(U_1), g^c(U_2))$ and $g(\bigcup_{i\in I} U_i) \succeq \max(\{g^c(U_i)\}_{i\in I})$. Note that $U_1 \cap U_2 \subseteq U_1$ and $U_1 \cap U_2 \subseteq U_2$, hence, by Lemma 8.4, $g^c(U_1 \cap U_2) \preceq g^c(U_1)$, $g^c(U_1 \cap U_2) \preceq g^c(U_2)$, therefore, $g(U_1 \cap U_2) \preceq \min(g^c(U_1), g^c(U_2))$. As $\forall i \in I : U_i \subseteq \bigcup_{i\in I} U_i$, $g^c(U_i) \preceq g^c(\bigcup_{i\in I} U_i)$. Therefore, $g(\bigcup_{i\in I} U_i) \succeq \max(\{g^c(U_i)\}_{i\in I})$. □

Lemma 8.6 (Truth lemma)**.** *For any $\varphi \in \mathcal{L}$ and $\Delta \in W^c$, $\varphi \in \Delta$ iff $M^c, \Delta \models \varphi$.*

Proof. By induction on φ. If $\varphi := p$ for some propositional variable $p \in Prop$, then the equivalence follows directly from the definition of V^c. Cases of Boolean connectives are straightforward. Assume that $\varphi := B_\alpha\psi$ for some $\psi \in \mathcal{L}$.

Left-to-right: assume that $B_\alpha\psi \in \Delta$. Then, $\widehat{\psi} \in \mathcal{E}^c_0$, and $g^c(\widehat{\psi}) = \alpha$. By our induction hypothesis, $\widehat{\psi} = [\![\psi]\!]_{M^c}$. Then, for any $U \in \tau^c$, if $U \subseteq [\![\psi]\!]_{M^c}$, then $U \subseteq \widehat{\psi}$. Then, by Lemma 8.4, for any $U \in \tau^c$, if $U \subseteq [\![\psi]\!]_{M^c}$, then $g^c(U) \preceq g^c(\widehat{\psi})$. So $\widehat{\psi}$ is the most reliable evidence in favour of ψ, and it is graded with α. Hence, $M^c, \Delta \models B_\alpha\psi$.

Right-to-left: assume that $M^c, \Delta \models B_\alpha\psi$. Then, there exists $U \in \tau^c$, such that $g^c(U) = \alpha$, $U \subseteq [\![\psi]\!]_{M^c}$, and for any $U' \in \tau^c$, if $U' \subseteq [\![\psi]\!]_{M^c}$, then $g^c(U') \preceq g^c(U)$. By the induction hypothesis, $[\![\psi]\!]_M = \widehat{\psi}$. We have two cases: either $U \in \mathcal{E}^c_0$ or not.

If $U \in \mathcal{E}^c_0$, then $U = \widehat{\rho}$ for some $B_\alpha\rho \in \Gamma$. As $\widehat{\rho} \subseteq \widehat{\psi}$, $\Box(\rho \to \psi) \in \Gamma$. From $\{B_\alpha\rho, \Box(\rho \to \psi)\} \subseteq \Gamma$, by the (Mon) axiom, it follows that $\bigvee_{\beta\succeq\alpha} B_\beta\psi \in \Gamma$. We claim that $B_\alpha\psi \in \Gamma$. Assume otherwise. Then, $\bigvee_{\beta\succ\alpha} B_\beta\psi \in \Gamma$. Let $\beta \in G$ be a grade, such that $\beta \succ \alpha$ and $B_\beta\psi \in \Gamma$. By def. of $\mathcal{E}^c_0$, $\widehat{\psi} \in \mathcal{E}^c_0$ and $g^c(\widehat{\psi}) = \beta$. As $\widehat{\psi} = [\![\psi]\!]_{M^c} \subseteq [\![\psi]\!]_{M^c}$, $\widehat{\psi}$ is a piece of evidence that supports ψ, and it is graded strictly higher than α. Therefore, $M^c, \Delta \not\models B_\alpha\psi$. Contradiction. Hence, $B_\alpha\psi \in \Gamma$.

If $U \notin \mathcal{E}^c_0$, then $g^c(U) = \alpha = \max(\{\alpha \in G \mid B_\alpha\rho \in \Gamma, \widehat{\rho} \subseteq U\})$. Let $\widehat{\rho} \in \mathcal{E}^c_0$ be a basic piece of evidence, such that $\widehat{\rho} \subseteq U$ and $g^c(\widehat{\rho}) = \alpha = \max(\{\alpha \in G \mid B_\alpha\rho \in \Gamma, \widehat{\rho} \subseteq U\})$. As $\widehat{\rho} \subseteq U \subseteq [\![\psi]\!]_{M^c}$, $B_\alpha\rho, \Box(\rho \to \psi) \in \Gamma$. By the (Mon) axiom, it follows that $\bigvee_{\beta\succeq\alpha} B_\beta\psi \in \Gamma$. Analogously to the previous case, we can demonstrate that if $\beta \succ \alpha$, then $B_\beta\psi \notin \Gamma$, as then $\widehat{\psi} \in \mathcal{E}^c_0$ would have been a piece of evidence that supports ψ and is graded strictly higher than α, contradicting $M^c, \Delta \models B_\alpha\psi$. Hence, $B_\alpha\psi$. □

Theorem 8.7 (Soundness and completeness w.r.t. **P**)**.** Given any set of formulae $\Gamma \cup \{\varphi\} \subseteq \mathcal{L}$, the following holds:

$$\Gamma \models_{\mathbf{P}} \varphi \Leftrightarrow \Gamma \vdash_{\mathbf{L}} \varphi$$

Proof. ($\Leftarrow$) By a routine check of the validity of the axioms and rules of inference.

($\Rightarrow$) By contraposition. Assume that $\Gamma \nvdash_{\mathbf{L}} \varphi$. Then, $\Gamma \cup \{\neg\varphi\}$ is **L**-consistent. Hence, there exists a maximal consistent set Δ, such that $\Gamma \cup \{\neg\varphi\} \subseteq \Delta$. Build a canonical model M^c, where $W^c = [\Delta]_\Box$. By Lemma 8.5, M^c is a pseudo-model. By Lemma 8.6, $\forall \psi \in \Gamma$, $M^c, \Delta \models \psi$ and $M^c, \Delta \models \neg\varphi$. Hence, $\Gamma \not\models_{\mathbf{P}} \varphi$. □

Next, we show how to transform any pseudo-model to a graded topo-e model that satisfies exactly the same formulae.

Let $M = (W, \mathcal{E}_0, \tau, g, V)$ be a pseudo-model. Let $X = \{U \in \tau \mid g(U) \neq \mathbf{0}, U \neq W\}$ be a collection of non-trivial opens that are not graded by $\mathbf{0}$. If $X = \emptyset$, then we can take a model $M' = (W, \mathcal{E}'_0, \tau', g', V)$, where $\mathcal{E}'_0 = \{W\}$, and $\tau' = \{\emptyset, W\}$ is a trivial topology. Consequently, $g'(\emptyset) = \mathbf{0}$, $g'(W) = \mathbf{1}$. Clearly, M' is a graded topo-e model.

Next, we show that $M, w \models \varphi$ iff $M', w \models \varphi$ for any $\varphi \in \mathcal{L}$ and any $w \in W$. We do it by induction on φ. Our induction hypothesis can be reformulated as follows: $[\![\varphi]\!]_M = [\![\varphi]\!]_{M'}$ for any $\varphi \in \mathcal{L}$. As the set of possible worlds and the valuation function in both models are the same, for any $\varphi \in \mathcal{L}$, if φ is modality-free, then $M, w \models \varphi$ iff $M', w \models \varphi$ for any $w \in W$.

Let $\varphi := B_\alpha\psi$. Assume that $M, w \models B_\alpha\psi$. $[\![\psi]\!]_M = W$ iff $M, w \models \Box\psi$. By induction hypothesis, $[\![\psi]\!]_{M'} = W$, so $M, w \models \Box\psi$ iff $M', w \models \Box\psi$. If $[\![\psi]\!]_M \subset W$, then, as $X = \emptyset$, $M, w \models B_{\mathbf{0}}\psi$. By induction hypothesis, $[\![\psi]\!]_M = [\![\psi]\!]_{M'} \subset W$, so there is no non-empty open set in τ' to support ψ. Hence, the most reliable piece of evidence to support ψ in M' is $\emptyset$. That is, $M', w \models B_{\mathbf{0}}\psi$. Analogously, since $\tau' = \{\emptyset, W\}$, for any formula ψ, either $M', w \models \Box\psi$ or $M', w \models B_{\mathbf{0}}\psi$. If $M', w \models B_{\mathbf{0}}\psi$, then $[\![\psi]\!]_{M'} = [\![\psi]\!]_M \neq W$, and as $X = \emptyset$, $M, w \models B_{\mathbf{0}}\psi$.

Next, we look into the non-trivial case, where $X \neq \emptyset$. First, we define a topo-e model $M' = (W', \mathcal{E}'_0, \tau', V')$ as follows:

1. $W' = W \times X$;
2. $\mathcal{E}'_0 = \{U \times \{U\} \mid U \in X\} \cup \{W'\}$;
3. τ' is a topology generated by $\mathcal{E}'_0$;
4. $V'(p) = \{(w, U) \in W' \mid w \in V(p)\}$;

In other words, we make isomorphic copies of every world. Every possible world $w \in W$ gets copies labelled by one of the non-trivial opens $U \in X$. Consequently, the basis of the topology consists of sets $\{(w, U) | w \in U\}$. It is clear that for any $U_1 \times \{U_1\}, U_2 \times \{U_2\} \in \mathcal{E}'_0$, If $U_1 \neq U_2$, then $U_1 \times \{U_1\} \cap U_2 \times \{U_2\} = \emptyset$: all worlds in $U_1 \times \{U_1\}$ are labelled with U_1, while all worlds in $U_2 \times \{U_2\}$ – with U_2. As the basis of the topology consists of mutually disjoint sets, $\mathcal{E}'_0 \cup \{\emptyset\}$ is already closed under taking intersections. Hence, $\tau' = \{\emptyset, W\} \cup \{\bigcup_{i\in I}(U_i \times \{U_i\}) \mid \{U_i\}_{i\in I} \subseteq X\}$.

Therefore, we the following grading function $g' : \tau' \to G$ is total and well-defined:

$$g'(O) = \begin{cases} \mathbf{1} & \text{if } O = W \\ \mathbf{0} & \text{if } O = \emptyset \\ \max(\{g(U_i)\}_{i\in I}) & \text{if } O = \bigcup_{i\in I}(U_i \times \{U_i\}) \text{ for some } \{U_i\}_{i\in I} \subseteq X \end{cases}$$

It is clear then that a tuple $M' = (W', \mathcal{E}'_0, \tau', g', V')$ is a graded topo-e model.

Next, we prove that $M, w \models \varphi$ iff $M', (w, U) \models \varphi$ for any $\varphi \in \mathcal{L}$ and $w \in W$, $(w, U) \in W'$.

Let $f : W' \to W$ be a function, such that $f((w, U)) = w$ for any $(w, U) \in W'$. Consequently, for any set of possible worlds $S' \subseteq W'$, $f[S'] = \{w \in W \mid \exists U \in X((w, U) \in S')\}$, and for any $S \subseteq W$, $f^{-1}[S] = \{(w, U) \in W' \mid w \in S\}$.

Note that for any $O \in \mathcal{E}'_0 \setminus \{W'\}$, $f[O] = U$ for some $U \in X$, such that $O = U \times \{U\}$. As $O = \{(w, U) \mid w \in U\}$, $f[O] = \{w \in W \mid \exists U \in X(w, U) \in O\} = U$. Consequently, for any $O \in \tau' \setminus \{\emptyset, W\}$, if $O = \bigcup_{i\in I} U_i \times \{U_i\}$ for some $\{U_i\}_{i\in I} \subseteq X$, $f[O] = \bigcup_{i\in I} U_i$, while $f[\emptyset] = \emptyset$, $f[W'] = W$. Therefore, $f : W' \to W$ is an open map.

Lemma 8.8. For any $O \in \tau'$ and any $S' \subseteq W'$, if $O \subseteq S'$, then $f[O] \subseteq f[S']$.

Proof. If $O = \emptyset$ or $O = W'$, then the result is trivial. Assume that $\emptyset \subset O \subset W'$. Then, $O = \bigcup_{i\in I}(U_i \times \{U_i\}) \subseteq S'$ for some $\{U_i\}_{i\in I}$. Consequently, $f[O] = \bigcup_{i\in I} U_i$. Take any $w \in \bigcup_{i\in I} U_i$. By definition, $(w, U_i) \in O$ for some U_i, such that $w \in U_i$. As $O \subseteq S'$, $(w, U_i) \in S'$. Therefore, there exists $U \in X$, such that $(w, U) \in S'$. Hence, $w \in f[S']$. □

Lemma 8.9. For every formula $\varphi \in \mathcal{L}$, $f^{-1}\big[f[[\![\varphi]\!]_{M'}]\big] = [\![\varphi]\!]_{M'}$.

Proof. Let us call a set $S' \subseteq W'$ *index-closed* iff for any $w \in W$ and any $U, U' \in X$, $(w, U) \in S'$ iff $(w, U') \in S'$. In other words, an index-closed set includes all isomorphic copies of any world that is inside it.

If $S' \subseteq W'$ is index-closed, then $W' \setminus S'$ is index-closed as well. For the contradiction, assume otherwise. So, there exists $(w, U) \in W' \setminus S'$ and an open $U' \in X$, such that $(w, U') \notin W' \setminus S'$. Then, $(w, U') \in S'$, while $(w, U) \notin S'$, which contradicts the assumption that S' is index-closed. If $S'_1, S'_2 \subseteq W'$ are index-closed, then $S'_1 \cup S'_2$ is index-closed as well. For the contradiction, assume otherwise. So there exists $(w, U) \in S'_1 \cup S'_2$ and an open $U' \in X$, such that $(w, U') \notin S'_1 \cup S'_2$. W.l.o.g., assume that $(w, U) \in S'_1$. Since $(w, U') \notin S'_1 \cup S'_2$, $(w, U') \notin S'_1$, which contradicts the assumption that S'_1 is index-closed.

Clearly, if $S' \subseteq W'$ is index-closed, then $f^{-1}\big[f[S']\big] = S'$. So we will prove that for any formula $\varphi \in \mathcal{L}$, $[\![\varphi]\!]_{M'}$ is index-closed. We will do it by induction on φ.

Base-case: $\varphi := p$ for some $p \in Prop$. $[\![p]\!]_{M'} = V'(p) = \{(w, U) \in W' \mid w \in V(p)\}$, which is index-closed.

$\varphi := \neg\psi$. Then, $[\![\neg\psi]\!]_{M'} = W' \setminus [\![\psi]\!]_{M'}$. By the induction hypothesis, $[\![\psi]\!]_{M'}$ is index-closed, hence, so is $W' \setminus [\![\psi]\!]_{M'}$.

$\varphi := \psi_1 \vee \psi_2$. Then, $[\![\psi_1 \vee \psi_2]\!]_{M'} = [\![\psi_1]\!]_{M'} \cup [\![\psi_2]\!]_{M'}$. By the induction hypothesis, both $[\![\psi_1]\!]_{M'}, [\![\psi_2]\!]_{M'}$ are index-closed, hence, so is $[\![\psi_1]\!]_{M'} \cup [\![\psi_2]\!]_{M'}$.

$\varphi := B_\alpha\psi$. Note that the truth condition for $B_\alpha\psi$ is world-independent:

$$M', (w, U) \models B_\alpha\psi \Leftrightarrow g(Int[\![\psi]\!]_{M'}) = \alpha$$

I.e., for any $(w, U), (w', U') \in W'$, $M', (w, U) \models B_\alpha\psi$ iff $M', (w', U') \models B_\alpha\psi$. Therefore, either $[\![B_\alpha\psi]\!]_{M'} = \emptyset$ or $[\![B_\alpha\psi]\!]_{M'} = W'$, while both $\emptyset$ and W' are index-closed.

Therefore, the truth-set $[\![\varphi]\!]_{M'}$ of every formula $\varphi \in \mathcal{L}$ is index-closed, so $f^{-1}\big[f[[\![\varphi]\!]_{M'}]\big] = [\![\varphi]\!]_{M'}$ for any $\varphi \in \mathcal{L}$. □

Corollary 8.9.1. When limited to the truth-sets of formulae, $f : \{[\![\varphi]\!]_{M'} \mid \varphi \in \mathcal{L}\} \to \{[\![\varphi]\!]_M \mid \varphi \in \mathcal{L}\}$ is bijective.

Theorem 8.10. For any formula $\varphi \in \mathcal{L}$, any world $w \in W$ and any open $U \in X$ of a pseudo-model M, the following holds:

$$M, w \models \varphi \Leftrightarrow M', (w, U) \models \varphi$$

or, equivalently:

$$[\![\varphi]\!]_M = f[[\![\varphi]\!]_{M'}] \text{ and } [\![\varphi]\!]_{M'} = f^{-1}[[\![\varphi]\!]_M].$$

Proof. We prove it by induction on φ. Our induction hypothesis: $[\![\varphi]\!]_M = f[[\![\varphi]\!]_{M'}]$ and $f^{-1}[[\![\varphi]\!]_M] = [\![\varphi]\!]_{M'}$.

Let $\varphi := p$ for some $p \in Prop$. Note that $[\![p]\!]_M = V(p)$ and $[\![p]\!]_{M'} = V'(p)$. By def. of V', $V'(p) = \{(w, U) \in W' \mid w' \in V(p)\}$, from which it follows that:

$$V(p) = \{w \in W \mid \exists U \in X((w, U) \in V'(p))\}.$$

Therefore, $f[[\![p]\!]_{M'}] = [\![p]\!]_M$ and $f^{-1}[[\![p]\!]_M] = [\![p]\!]_{M'}$.

Cases of Boolean connectives are trivial.

So it is left for us to show that $M', (w, U) \models B_\alpha\varphi$ iff $M, f((w, U)) \models B_\alpha\varphi$.

Left-to-right: Assume that $M, (w, U) \models B_\alpha\varphi$. Then, there exists $O \in \tau'$, such that $O \subseteq [\![\varphi]\!]_{M'}$, $g'(O) = \alpha$ and for any $O' \in \tau'$, if $O' \subseteq [\![\varphi]\!]_{M'}$, then $g'(O') \preceq \alpha$. We have three cases: either $O = \emptyset$, or $O = W'$ or $O = \bigcup_{i\in I}(U_i \times \{U_i\})$ for some $\{U_i\}_{i\in I} \subseteq X$.

Case 1: $O = \emptyset$. Then, $g'(O) = \mathbf{0}$, and $M', (w, U) \models B_\mathbf{0}\varphi$. For the sake of contradiction, assume that $M, w \not\models B_\mathbf{0}\varphi$. Then, there exists an open $U \in \tau$, such that $g(U) \neq \mathbf{0}$ and $U \subseteq [\![\varphi]\!]_M$. So either $U = W$ or $U \in X$. If the former, then $[\![\varphi]\!]_M = W = [\![\top]\!]_M$. By IH, $f[[\![\varphi]\!]_{M'}] = [\![\varphi]\!]_M$ and $f[[\![\top]\!]_{M'}] = [\![\top]\!]_M$. So $[\![\varphi]\!]_M = [\![\top]\!]_M$ entails $f[[\![\varphi]\!]_{M'}] = f[[\![\top]\!]_{M'}]$. As f, when restricted to the truth-sets, is bijective, $[\![\varphi]\!]_{M'} = [\![\top]\!]_{M'} = W'$. In that case, $M', (w, U) \models \Box\varphi$, contradicting the initial assumption that $M', (w, U) \models B_\mathbf{0}\varphi$. So, assume the latter: $U \in X$. Then, $g(U) = \alpha$ for some $\alpha \in G \setminus \{\mathbf{0}, \mathbf{1}\}$, and $U \times \{U\} \in \tau'$, and $U \subseteq [\![\varphi]\!]_M$. By IH, $[\![\varphi]\!]_{M'} = f^{-1}[[\![\varphi]\!]_M]$, and as $U \subseteq [\![\psi]\!]_M$, for every $w \in U$, as $(w, U) \in f^{-1}[[\![\varphi]\!]_M]$, $M', (w, U') \models \varphi$. Therefore, $U \times \{U\} \subseteq [\![\psi]\!]_{M'}$, while $U \times U \in \tau$, and $g(U \times \{U\}) \succ \mathbf{0}$,

contradicting $M', (w, U) \models B_{\mathbf{0}}\psi$. Therefore, there can be no non-$\mathbf{0}$-graded opens $U \in \tau$, such that $U \subseteq [\![\psi]\!]_M$, i.e., $M, w \models B_{\mathbf{0}}\psi$.

Case 2: $O = W$. Then, $M', (w, U) \models \Box\psi$. Then, $[\![\psi]\!]_{M'} = W'$. By IH, $[\![\psi]\!]_M = f[[\![\psi]\!]_{M'}] = f[W'] = W$, i.e., $M, w \models \Box\psi$.

Case 3: $O = \bigcup_{i\in I}(U_i \times \{U_i\}) \neq W$ for some $\{U_i\}_{i\in I} \subseteq X$. As, by our initial assumption, $O \neq W$, $\alpha \neq \mathbf{1}$. Then, $g'(O) = \alpha = \max(\{g(U_i)\}_{i\in I})$. W.l.o.g., let $U_j \in \{U_i\}_{i\in I}$ be an open, such that $g(U_j) = \alpha$. By IH, $[\![\varphi]\!]_M = f[[\![\varphi]\!]_{M'}]$. By Lemma 8.8, as $O \subseteq [\![\varphi]\!]_{M'}$, $f[O] = \bigcup_{i\in I} U_i \subseteq f[[\![\varphi]\!]_{M'}] = [\![\psi]\!]_M$. As $U_j \subseteq \bigcup_{i\in I} U_i$, $U_j \subseteq [\![\varphi]\!]_M$. For the contradiction, assume that $M, w \not\models B_\alpha\varphi$. As $g(U_j) = \alpha$ and $U_j \subseteq [\![\varphi]\!]_M$, $M, w \models B_\beta\varphi$ for some $\beta \in G$, s.t. $\beta \succ \alpha$. Then, there exists an open $U_l \in \tau$, such that $g(U_l) = \beta$ and $U_l \subseteq [\![\varphi]\!]_M$. Either $U_l = W$ or $U_l \in X$. If $U_l = W$, then, $[\![\varphi]\!]_M = [\![\top]\!]_M = W$, so, as we have shown before, $W' = [\![\varphi]\!]_{M'}$. But then, $M', (w, U) \models \Box\varphi$, contradicting our initial assumption that $O \neq W$. Therefore, $U_l \in X$. So $U_l \times \{U_l\} \in \tau'$, and $g'(U_l \times \{U_l\}) = \beta$. As $U_l \subseteq [\![\varphi]\!]_M$, by IH, $f^{-1}[U_l] = U_l \times \{U_l\} \subseteq [\![\psi]\!]_{M'}$, while $g'(U_l \times \{U_l\}) = \beta$, $\beta \succ \alpha$, which contradicts $M', (w, U) \models B_\alpha\psi$. Hence, U_j is the most reliable piece of evidence to support ψ in M, i.e., $M, w \models B_\alpha\psi$.

Right-to-left: Assume that $M, w \models B_\alpha\varphi$. We have three cases: either $\alpha = 0$ or $\alpha = \mathbf{1}$ or $\mathbf{0} \prec \alpha \prec \mathbf{1}$.

If $M, w \models B_{\mathbf{0}}\varphi$, then $g(Int([\![\varphi]\!]_M)) = \mathbf{0}$. As M is a pseudo-model, $Int([\![\varphi]\!]_M)$ might be non-empty, despite it being graded with $\mathbf{0}$. For the contradiction, assume that $M', (w, U) \not\models B_{\mathbf{0}}\varphi$. Then, there exists a non-empty open $O \in \tau'$, such that $O \subseteq [\![\varphi]\!]_{M'}$. By Lemma 8.8, $f[O] \subseteq f[[\![\varphi]\!]_{M'}]$, and by IH, $f[O] \subseteq [\![\varphi]\!]_M$. As f is an open map, $f[O] \in \tau$. If $O = W'$, then $f[O] = W$, so $W \subseteq [\![\varphi]\!]_M$, i.e., $M, w \models \Box\varphi$, contradicting the initial assumption that $M, w \models B_{\mathbf{0}}\varphi$. So, assume that $O = \bigcup_{i\in I}(U_i \times \{U_i\}) \neq W'$ for some $\{U_i\}_{i\in I} \subseteq X$. In that case, $f[O] = \bigcup_{i\in I} U_i$ for some $\{U_i\}_{i\in I} \subseteq X$. As $\{U_i\}_{i\in I} \subseteq X \subseteq \tau$, $\bigcup_{i\in I} U_i \in \tau$. As every $U_i \in X$, $g(U_i) \succ \mathbf{0}$. As $U_i \subseteq \bigcup_{i\in I} U_i$, $g(\bigcup_{i\in I} U_i) \succeq g(U_i) \succ \mathbf{0}$. I.e., there exists an open set $\bigcup_{i\in I} U_i \in \tau$, such that $\bigcup_{i\in I} U_i \subseteq [\![\varphi]\!]_M$, and this open is graded strictly higher than $\mathbf{0}$, which contradicts the assumption that $M, w \models B_{\mathbf{0}}\varphi$. Hence, $M', (w, U) \models B_{\mathbf{0}}\varphi$.

If $M, w \models \Box\varphi$, then $W = [\![\varphi]\!]_M$. By IH, $f^{-1}[[\![\varphi]\!]_M] = [\![\varphi]\!]'_M$, and as $f^{-1}[W] = W'$, $[\![\varphi]\!]_{M'} = W'$. Hence, $W', (w, U) \models \Box\varphi$.

Finally, assume that $M, w \models B_\alpha\varphi$ for some $\mathbf{0} \prec \alpha \prec \mathbf{1}$. Then, $g(Int([\![\varphi]\!]_M)) = \alpha$. Let $Int([\![\varphi]\!]_M = U_\varphi \in \tau$. As $g(U_\varphi) \prec \mathbf{1}$, $U_\varphi \neq W$. Then, as $g(U_\varphi) \succ \mathbf{0}$, $U_\varphi \in X$. For the contradiction, assume that $M', (w, U) \not\models B_\alpha\varphi$. As $U_\varphi \in X$, $U_\varphi \times \{U_\varphi\} \in \tau'$, and $g'(U_\varphi \times \{U_\varphi\}) = \alpha$. By IH, $[\![\varphi]\!]_{M'} = \{(w, U) \in W' \mid w \in [\![\varphi]\!]_M\}$. As $U_\varphi \subseteq [\![\varphi]\!]_M$, $U_\varphi \times \{U_\varphi\} \subseteq [\![\varphi]\!]_{M'}$. Therefore, $M', (w, U) \models \bigvee_{\beta\succeq\alpha} B_\beta\varphi$. For the contradiction, assume that $M', (w, U) \models B_\beta\varphi$ for some $\beta \in G$, such that $\beta \succ \alpha$. Then, there exists an open $O \in \tau'$, s.t. $g'(O) = \beta$ and $O \subseteq [\![\varphi]\!]_{M'}$. Then, by Lemma 8.8, $f[O] \subseteq f[[\![\varphi]\!]_{M'}]$. By IH, $f[O] \subseteq [\![\varphi]\!]_M$. As f is an open map, $f[O] \in \tau$. If $O = W'$, then $f[O] = W$, but then $M, w \models \Box\varphi$, contradicting that $M, w \models B_\alpha\varphi$ for some $\mathbf{0} \prec \alpha \prec \mathbf{1}$. If $O = \bigcup_{i\in I}(U_i \times \{U_i\})$ for some $\{U_i\}_{i\in I} \subseteq X$, then $f[O] = \bigcup_{i\in I} U_i \in \tau$. As $g'(O) = \max(g(U_i) \mid i \in I)$, and $U_i \subseteq \bigcup_{i\in I} U_i$ for every $i \in I$, $g(\bigcup_{i\in I} U_i) \succeq g'(O) = \beta \succ \alpha$. Therefore, there exists an open $\bigcup_{i\in I} U_i \in \tau$, graded strictly higher than α, such that $\bigcup_{i\in I} U_i \subseteq [\![\varphi]\!]_M$, which contradicts that $M, w \models B_\alpha\varphi$ for some $\mathbf{0} \prec \alpha \prec \mathbf{1}$. Therefore, $M', (w, U) \models B_\alpha\varphi$.

□

Theorem 8.11. **L** is sound and strongly complete w.r.t. the class of graded topo-e models. In other words, for every set of formulae $\Gamma \subseteq \mathcal{L}$ and any formulae $\varphi \in \mathcal{L}$, the following holds:

$$\Gamma \vdash_{\mathbf{L}} \varphi \Leftrightarrow \Gamma \models \varphi$$

Proof. ($\Rightarrow$) By routine checking, the axioms and rules of inference are valid.

($\Leftarrow$) By contraposition. Assume that $\Gamma \nvdash_{\mathbf{L}} \varphi$. Then, $\Gamma \cup \{\neg\varphi\}$ is **L**-consistent. By the Theorem 8.7, there exists a pseudo-model $M = (W, \mathcal{E}_0, \tau, g, V)$ and a world $w \in W$, such that $M, w \models \Gamma \cup \{\neg\varphi\}$. By Theorem 8.10, there exists a graded topo-e model $M' = (W', \mathcal{E}'_0, \tau', g', V')$ and a world $(w, U) \in W'$, such that $M', (w, U) \models \Gamma \cup \{\neg\varphi\}$. Therefore, $\Gamma \not\models \varphi$. □

References

Balinski, Michel and Rida Laraki (2007). "A theory of measuring, electing, and ranking". In: *Proceedings of the National Academy of Sciences* 104.21, pp. 8720–8725.

— (2011). *Majority judgment: measuring, ranking, and electing*. The MIT Press.

Baltag, Alexandru, Nick Bezhanishvili, et al. (2016). "Justified belief and the topology of evidence". In: *International Workshop on Logic, Language, Information, and Computation*. Springer, pp. 83–103.

— (2022). "Justified belief, knowledge, and the topology of evidence". In: *Synthese* 200.6, p. 512.

Baltag, Alexandru and Sonja Smets (2006). "Conditional doxastic models: A qualitative approach to dynamic belief revision". In: *Electronic notes in theoretical computer science* 165, pp. 5–21.

Fagin, Ronald et al. (2004). *Reasoning about knowledge*. MIT press.
Halpern, Joseph Y (1997). "Defining relative likelihood in partially-ordered preferential structures". In: *Journal of Artificial Intelligence Research* 7, pp. 1–24.
Hintikka, Jaakko (1962). *Knowledge and Belief: An Introduction to the Logic of the Two Notions*. English. Ithaca, N.Y.: Cornell University Press.
Jeffrey, Richard C (1990). *The logic of decision*. University of Chicago press.
Kolmogorov, A. N. (1950). *Foundations of the Theory of Probability*. New York: Chelsea Publishing Company.
Kyburg Henry E., Jr. (1961). *Probability and the Logic of Rational Belief*. Middletown, CT: Wesleyan University Press.
Lehrer, Keith (1999). "Justification, coherence and knowledge". In: *Erkenntnis (1975-)* 50.2/3, pp. 243–258.
Leitgeb, Hannes (2014). "The stability theory of belief". In: *Philosophical review* 123.2, pp. 131–171.
Lyon, Aidan and Michael Morreau (2018). "The wisdom of collective grading and the effects of epistemic and semantic diversity". In: *Theory and Decision* 85.1, pp. 99–116.
Makinson, David C (1965). "The paradox of the preface". In: *Analysis* 25.6, pp. 205–207.
McKinsey, John Charles Chenoweth and Alfred Tarski (1944). "The algebra of topology". In: *Annals of mathematics* 45.1, pp. 141–191.
Ramsey, Frank P (1926). "Truth and probability". In: *Readings in formal epistemology: Sourcebook*. Springer, pp. 21–45.
Savage, Leonard J (2012). *The foundations of statistics*. Courier Corporation.
Segerberg, Krister (1971). "Qualitative probability in a modal setting". In: *Studies in Logic and the Foundations of Mathematics*. Vol. 63. Elsevier, pp. 341–352.
Van Benthem, Johan and Eric Pacuit (2011). "Dynamic logics of evidence-based beliefs". In: *Studia Logica* 99.1, p. 61.
Van Ditmarsch, Hans et al. (2015). *Handbook of epistemic logic*. College Publications.